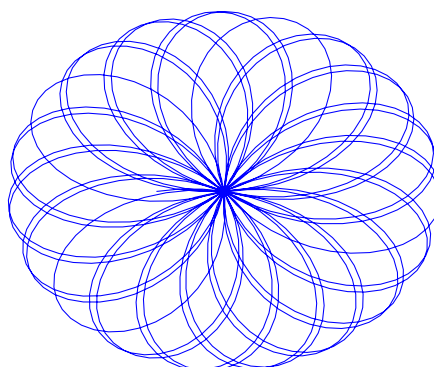
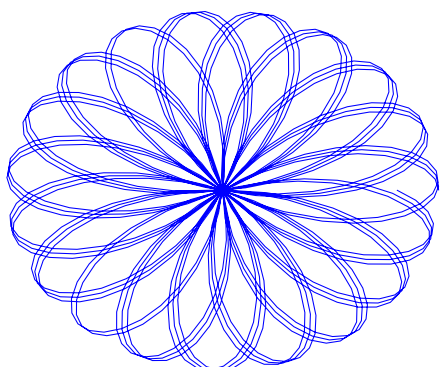

Beginner's Guide to PGFPlots

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Beginner's Guide to PGFPlots

1.1 Motivation

When writing scientific documents, it is often desirable to produce plots that match the fonts, style, and formatting of the document itself. External plotting tools such as Python, MATLAB, or GNUplot can generate figures, but they often require importing external images.

1.2 What is PGFPlots?

PGFPlots is a powerful LaTeX package which allows us to generate high-quality scientific plots directly inside LaTeX. It is built on top of the powerful graphics framework *TikZ*. Because *PGFPlots* uses the same typesetting engine as LaTeX, all labels, mathematical expressions, and fonts appear consistent with the rest of the document.

Using *PGFPlots* we can easily create

- (a) function plots
- (b) scatter plots
- (c) parametric curves
- (d) surface plots
- (e) data visualizations

while maintaining full control over labels, fonts, and styling.

1.3 Loading the Package

To use PGFPlots, include the following in the preamble of your document:

```
1 \usepackage{pgfplots}
2 \pgfplotsset{compat=1.18} % or the latest version you have
```

The `compat` option ensures that the behaviour of the package remains consistent across versions.

1.4 Your First PGFPlots Figure

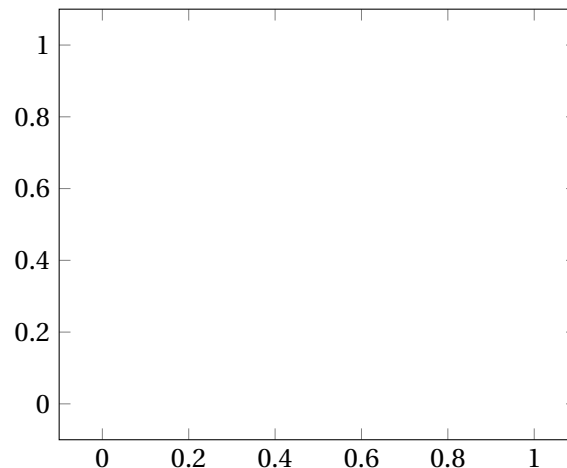
The basic environment used for plotting is the axis environment inside a *tikzpicture*. As it is built on top of TikZ, we first need to load the environment in the preamble.

```
1 \usepackage{tikz}
```

Here is a very basic code which will be the backbone of your plot.

```
1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}
4 \end{axis}
5 \end{tikzpicture}
6 \end{center}
```

You will see something like this. Okay, It has created a box on which we can plot things.



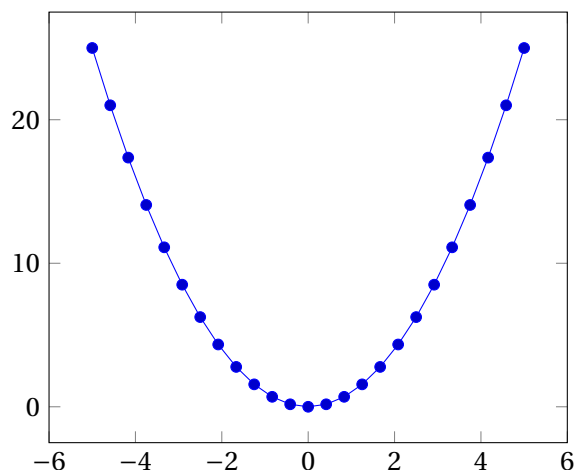
Let's pause and explain what it does. “center” environment aligns the plot to the center. “tikzpicture” is the TikZ environment which creates drawing canvas. “Axis” environment defines the coordinate system and axes. Inside this we will be plotting functions, data etc.

1.5 Plotting a Function & Basic Syntax

First let's try to plot a very basic function $f(x) = x^2$.

```
1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}
4 \addplot {x^2};
5 \end{axis}
6 \end{tikzpicture}
7 \end{center}
```

This code produces the plot of the function over a default domain which is $[-5, 5]$ with a total 25 no of sample points.



“`addplot`” command draws the function or data. The basic syntax is `\addplot[options]{function/data};`. Let us explore the options.

(a) `\addplot[domain=-2:2]{x^2};` → “domain” changes the range to $[-2, 2]$.

(b) `\addplot[samples=50]{x^2};` → “no of sample points” changes to 50.

(c) `\addplot[red,dotted,thick]{x^2};` → changes the colour to “red” joined by “dotted” lines which are “thick”. Colors can be “rgb” or mixture of this as you prefer, like “red!20!blue” means 20% red and 80% blue. The line style options are “solid(by default), dashed, dotted, dashdotted, densely dashed, loosely dashed, densely dotted, loosely dotted” etc. The line width options are “ultra thin, very thin, thin, semithick, thick, very thick, ultra thick”.

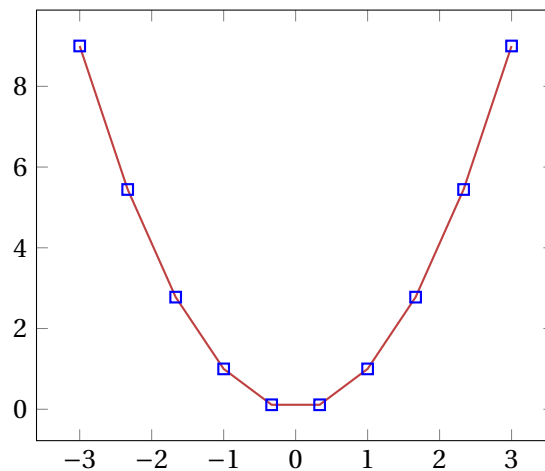
(d) `\addplot[mark=*]{x^2};` → sets the point marker to “*”. Other options are “o,triangle,square” etc. “*” which is filled circle, “o” is the unfilled one.

You can combine them to make the plot as you desire. Here is a minimal example of this :

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}
4 \addplot[domain=-3:3,
5 samples=10,
6 red!50!gray,thick,
7 mark=square,mark options={color=blue}] {x^2};
8 \end{axis}
9 \end{tikzpicture}
10 \end{center}

```



The line mark options={color=blue} changes the marker colour to blue. By default the marker colour is the same colour as “line colour”.

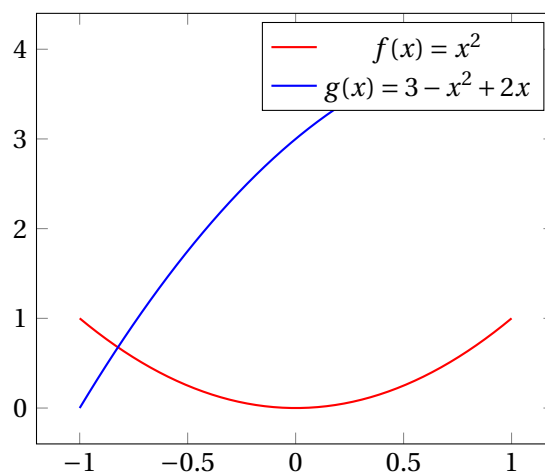
1.6 Add a Legend

So far we have discussed how to plot a function. The next thing you might wanna do is to add a “legend” to the plot. This is very useful when in one figure you plot many functions. Consider you want to plot two functions $f(x) = x^2$ and $g(x) = 3 - x^2 + 2x$. Also you want to label which plot is which. Here is a minimal example that shows you how to do that.

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}
4 \addplot[domain=-1:1,samples=50,red,thick]{x^2};
5 \addlegendentry{$f(x)=x^2$}
6 \addplot[domain=-1:1,samples=50,blue,thick]{3-x^2+2*x};
7 \addlegendentry{$g(x)=3-x^2+2*x$}
8 \end{axis}
9 \end{tikzpicture}
10 \end{center}

```



Let's analyse the lines :

- (a) `\addplot[domain=-1:1,samples=50,red,thick]{x^2}`; makes the plot of $f(x) = x^2$, in range $[-1, 1]$ with 50 samples in this range and connects them with solid red thick line.
- (b) `\addplot[domain=-1:1,samples=50,blue,thick]{3-x^2+2*x}`; makes the plot of $g(x) = 3 - x^2 + 2x$, in range $[-1, 1]$ with 50 samples in this range and connects them with solid blue thick line.
- (c) In order to label your plot you have to use `\addlegendentry{your plot label}`. Note that there is no semi-colon at the end of this line.
- (d) So the lines `\addlegendentry{$f(x)=x^2$}` and `\addlegendentry{$g(x)=3-x^2+2x$}` add legends to the functions respectively. This creates a legend box at the “north east” corner of the figure. It is default set. For now don't worry, we will come to this later on.

So in short “`\addlegendentry{your plot label}`” is the command using which you can add the legend.

Interesting Comment:

You can add legends to all your functions using this command: “`\legend{legend1, legend2, ...}`”. In this case this becomes “`\legend{$f(x)=x^2$, $g(x)=3-x^2+2x$}`”. Try this one.

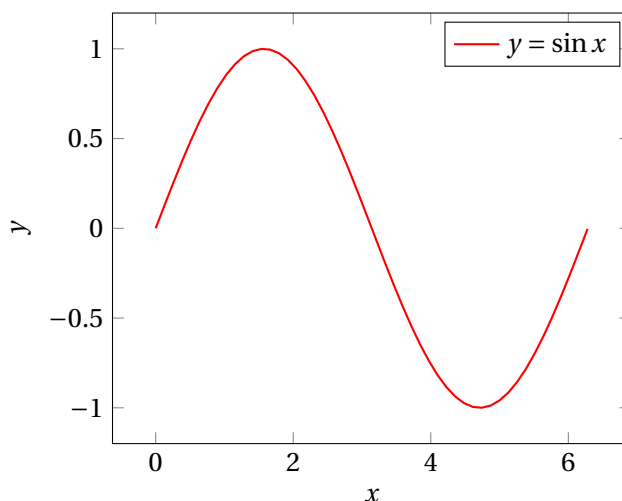
1.7 Label Axes

We will now move on how to “label” the axes. Consider you want to plot $y = \sin x$, where you want to label the axes as “ x ” and “ y ”. You can do this by controlling the “axis” environment. Here is how you can do it :

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[xlabel={x},ylabel={y}]
4 \addplot[domain=0:6.28,samples=50,red,thick]{sin(deg(x))};
5 \addlegendentry{$y=\sin{x}$}
6 \end{axis}
7 \end{tikzpicture}
8 \end{center}

```



Let's analyse :

- (a) The axis options “`[xlabel={ $x$ }, ylabel={ $y$ }]`” adds the labels x to x axis and $y = \sin x$ to y axis respectively. There is no need to put `\{...\}` braces, but we put it here. By this, I mean “`[xlabel= $x$ , ylabel= $y$ ]`” will also results in same plot.
- (b) The line “`\addplot[domain=0:6.28,samples=50,red,thick]{sin(deg(x))};`” plots the function $\sin x$ in the domain $[0, 2\pi]$. Notice that we have put “`{sin(deg(x))}`”, a “deg()” to signify input is in radians.
- (c) “`\addplot[domain=0:360,samples=50,red,thick]{sin(x)};`” is the equivalent command which takes the variable in degrees.
- (d) “`\addlegendentry{ $y = \sin\{x\}$ }`” adds the legend.
- (e) Now looking at the plot, you can very easily see that “ylabel” is along the vertical. You might want to put it along the horizontal. So we will use the rotate command to make it “horizontal”. The command is “`ylabel style={rotate=rotate angle}`”. For example to rotate “ y label” by 270° clockwise we use: `[xlabel={ $x$ }, ylabel={ $y$ }, ylabel style={rotate=270}]` command.
- (f) Sometimes you might want to shift the labels by specific amount. By default it is place at the middle of the axis. For example `xlabel style={xshift=10pt, yshift = 5pt, rotate = 45}` shifts the label along x by 10pt, along y 5 pt and performs a clockwise rotation of 45° . You can do the same for y axis also by using `ylabel style={place your operations here}`.

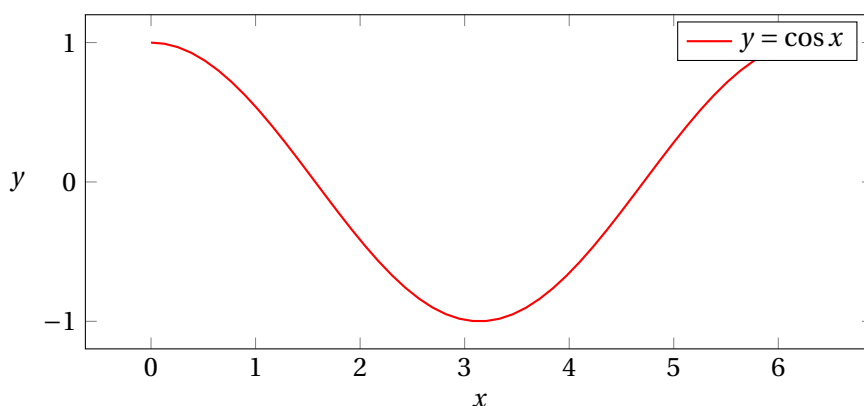
1.8 Controlling Height and Width of the Figure

The size of a plot can be controlled using the width and height options inside the axis environment. These options determine the physical size of the plot in the document.

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[xlabel={ $x$ },ylabel={ $y$ },ylabel style={rotate=270},width=12cm,height=6cm]
4 \addplot[domain=0:6.28,samples=50,red,thick]{cos(deg(x))};
5 \addlegendentry{ $y = \cos\{x\}$ }
6 \end{axis}
7 \end{tikzpicture}
8 \end{center}

```



- (a) There are two options “height” and “width”. We can set them as we desire to. They can be expressed in “cm, mm, in, \linewidth, \textwidth” units. The syntax is `width=<length>, height=<length>`.
- (b) `width=12cm, height=6cm` creates a figure that is 12 cm wide and 6 cm tall.
- (c) Often it is convenient to scale the plot relative to the page width. For example, height and width can be set this way “`width=0.7\textwidth, height=0.4\textwidth`”. This makes the figure adapt automatically to the document layout.
- (d) Changing width and height does not distort the plot, PGFPlots automatically rescales the axis.

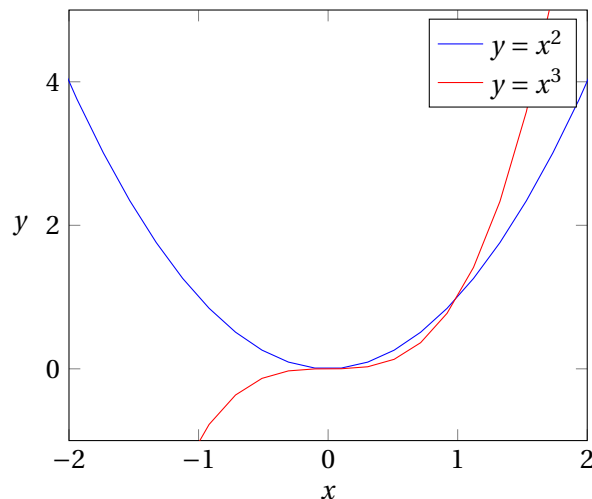
1.9 Modifying Axis Limits

The visible range of the plot is controlled by specifying the limits of the axes. This is useful when we want to focus on a particular region of a function or restrict the displayed interval. A basic code is shown below :

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[xlabel={$x$},ylabel={$y$},ylabel style={rotate=270},
4 xmin=-2,xmax=2,ymin=-1,ymax=5]
5 \addplot[domain=-5:5,samples=50,blue,thin]{x^2};
6 \addlegendentry{$y=x^2$}
7 \addplot[domain=-5:5,samples=50,red,thin]{x^3};
8 \addlegendentry{$y=x^3$}
9 \end{axis}
10 \end{tikzpicture}
11 \end{center}

```



- (a) `\addplot[domain=-5:5,samples=50,blue,thin]{x^2};` plots x^2 and `\addplot[domain=-5:5,samples=50,red,thin]{x^3};` plots x^3 in the domain $[-5,5]$. But we want to focus only on the range say $-2 \leq x \leq 2$ and $-1 \leq y \leq 5$.
- (b) The basic syntax for setting the limits is `xmin=<value>, xmax=<value>, ymin=<value>, ymax=<value>` inside the axis environment.
- (c) As shown before `xmin=-2, xmax=2, ymin=-1, ymax=5` restricts the values as we desired.

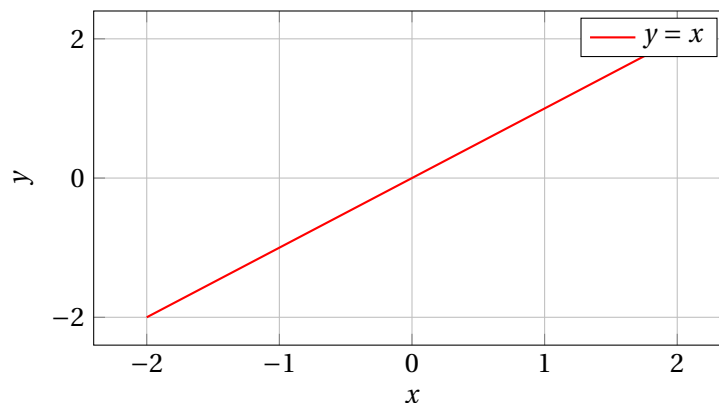
1.10 Add Grid and Change Grid Style

Grids are very useful for improving the readability of plots. They help the reader estimate values directly from the graph. A grid can be added using the `grid` option inside the axis environment. The simplest way to add a grid is by writing `grid=major` in the axis environment.

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[xlabel=$x$,ylabel=$y$,grid=major,height=6cm,width=10cm]
4 \addplot[domain=-2:2,samples=50,red,thick] {x};
5 \addlegendentry{$y=x$};
6 \end{axis}
7 \end{tikzpicture}
8 \end{center}

```



Let's describe what's happening :

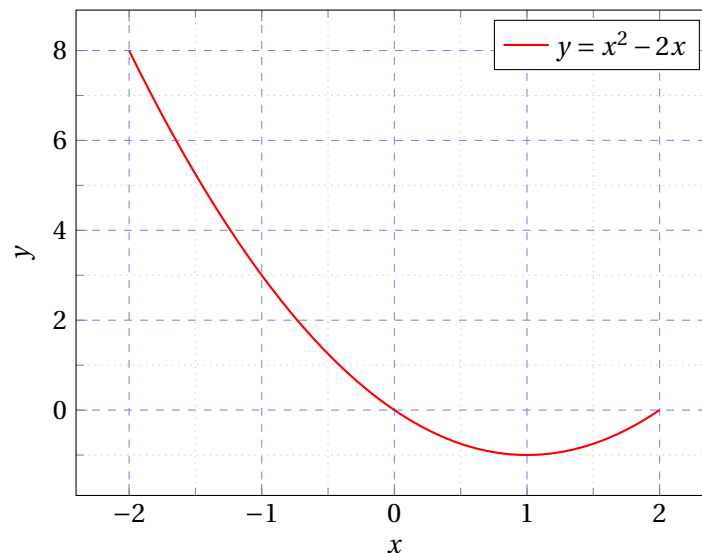
- (a) `grid=major` draws grid lines at the major tick positions.
- (b) The options for `grid=` options are “major, minor, both”.
- (c) To use minor grids, we must also define minor ticks. For example `minor tick num=1` places one minor grid in between two major grid points.

Sometimes we might want to change the grid styles. The appearance of the grid can be modified using `major grid style={...}` and `minor grid style={...}` commands. We can specify the line style and colour of the grid lines as desired. For example `major grid style={blue,very thin}` creates solid blue coloured very thin major grid lines and `minor grid style={dashed,gray}` creates dashed gray coloured minor grid lines. Here is an example to show it :

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[xlabel=$x$,ylabel=$y$,grid=both,
4 major grid style={dashed, blue!50},minor grid style={dotted, gray!50},
5 minor tick num=1,height=8cm,width=10cm]
6 \addplot[domain=-2:2,samples=50,red,thick] {x^2-2*x};
7 \addlegendentry{$y=x^2-2x$};
8 \end{axis}
9 \end{tikzpicture}
10 \end{center}

```



1. `\addplot[domain=-2:2,samples=50,red,thick]{x^2-2*x};` draws the function $x^2 - 2x$ in the range $[-2, 2]$ with a red thick line.
2. `grid=both` draws both grids.
3. `minor tick num=1` places one minor tick in between two major ticks.
4. `major grid style={dashed, blue!50}` defines the pattern for major grid, which sets it to be dashed style and a colour having 50% blue and 50% white.
5. `minor grid style={dotted, gray!50}` defines the pattern for minor grid, which sets it to be dotted style and a colour having 50% gray and 50% white.

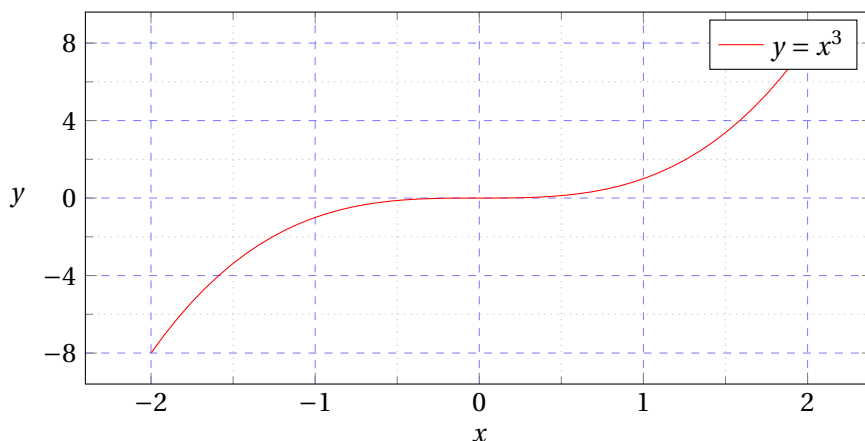
1.11 Add Ticks to axes

Ticks are the small marks on the axes that indicate numerical values. By default, PGFPlots automatically chooses suitable tick positions. Using `xtick={<values>}, ytick={<values>}` command in the axis environment we can specify them manually. For example `xtick={-2, -1, 0, 1, 2}, ytick={0, 1, 2, 3, 4}` can be used to set ticks on x and y axes respectively.

```

1  \begin{center}
2  \begin{tikzpicture}
3  \begin{axis}[xlabel={\$x\$},ylabel={\$y\$},ylabel style={rotate=270},grid=both,
4  major grid style={dashed, blue!50},minor grid style={dotted, gray!50},
5  minor tick num=1,height=6.5cm,width=12cm,
6  xtick={-2,-1,...,2},ytick={-8,-4,...,8}]
7  \addplot[domain=-2:2,samples=50,red,thin]{x^3};
8  \addlegendentry{\$y=x^3\$}
9  \end{axis}
10 \end{tikzpicture}
11 \end{center}

```



- (a) `\addplot[domain=-2:2,samples=50,red,thin]{x^3}`; plots the function x^3 in the range $[-2,2]$.
- (b) `xtick={-2,-1,...,2}` sets ticks along x axis. The “...” are used to simplify the ticks. PGFPlots automatically fills the intermediate values.
- (c) `ytick={0,1,...,4}` is the same command as described for x .

Interesting Comment:

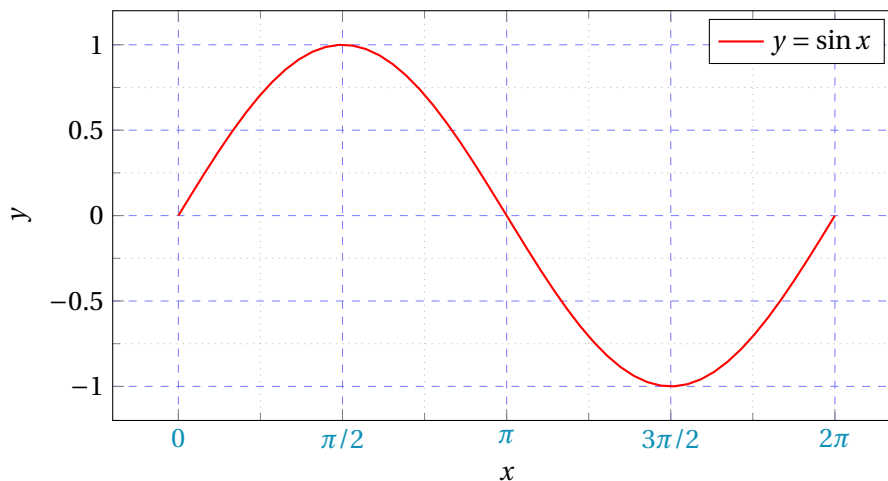
If you want no ticks, you can disable them using `xtick=\empty`, `ytick=\empty`. We can set the tick positions automatically using `xtick distance=<distance>`, `ytick distance=<distance>`.

Sometimes we also want to write some custom texts at the tick positions. This can be achieved by this command `xtick={tick position},xticklabels={tick labels}`.

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[xlabel={x},ylabel={y},grid=both,
4 major grid style={dashed, blue!50},minor grid style={dotted, gray!50},
5 minor tick num=1, xtick={0,90,180,270,360},width = 12cm,height = 7cm,
6 xticklabels={0$, \pi/2$, \pi$, 3\pi/2$, 2\pi$}, xticklabel style={cyan!70!black}]
7 \addplot[domain=0:360,samples=50,red,thick]{sin(x)};
8 \addlegendentry{$y=\sin{x}$}
9 \end{axis}
10 \end{tikzpicture}
11 \end{center}

```



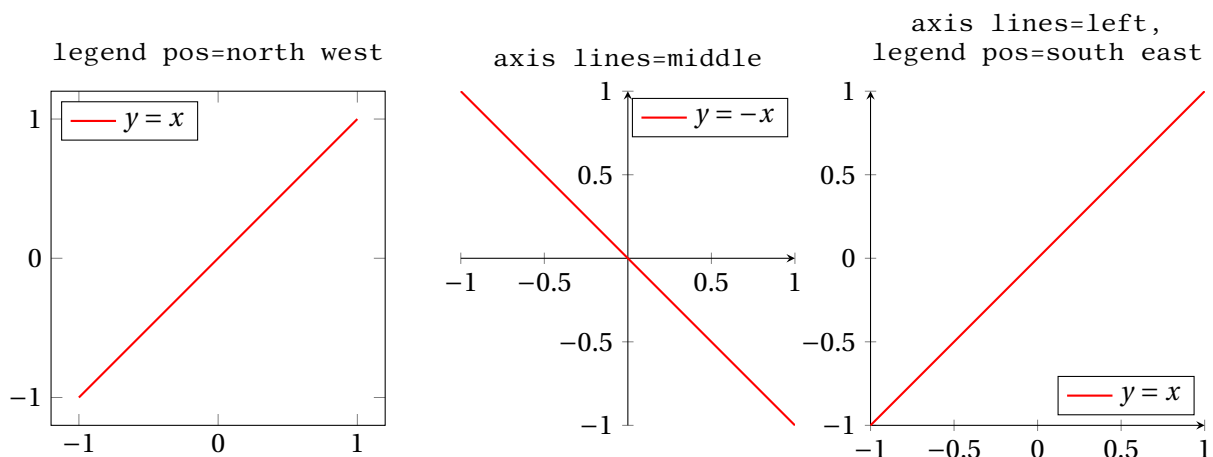
- (a) `xtick={0, 90, 180, 270, 360}` sets the tick positions at those respective angles.
- (b) `xticklabels={\$0\$, \$\pi/2\$, \$\pi\$, \$3\pi/2\$, \$2\pi\$}` sets the ticklabels.
- (c) Similarly we can do for the *y* axis also using `ytick={...}`, `yticklabels={...}`.
- (d) As we did for the `xlabel` style we can use `xticklabel` style the same way.
- (e) For example `yticklabel style={rotate=45}` will rotate the “yticks” by 45°.

1.12 Additional Options

The axis environment in PGFPlots provides many options that control the appearance and behavior of the plot. In this section we discuss some commonly used options such as axis lines, legend position, axis line style, and equal axis scaling.

- Axis lines:** By default, the axes appear as a box surrounding the plot. However, we can change how the axes are drawn using the `axis lines` option. Common settings include “`axis lines=box`, (default set), `axis lines=left`, `axis lines=middle`, `axis lines=center`”. If you don’t want the axes to be drawn then use `axis lines = none`.
- Legend position:** When multiple plots are present, we can place the legend using the `legend pos` option. Common options are “`legend pos=north east`, `legend pos=north west`, `legend pos=south east`, `legend pos=south west`”. `legend style={at={(x,y)}, anchor=...}` gives you full control. For example `legend style={at={(0.5,0.95)}, anchor=north}` adds the legend at (0.5,0.95) and anchors it to north.

See the following figure that explores different “Axis lines” options and “Legend position”.



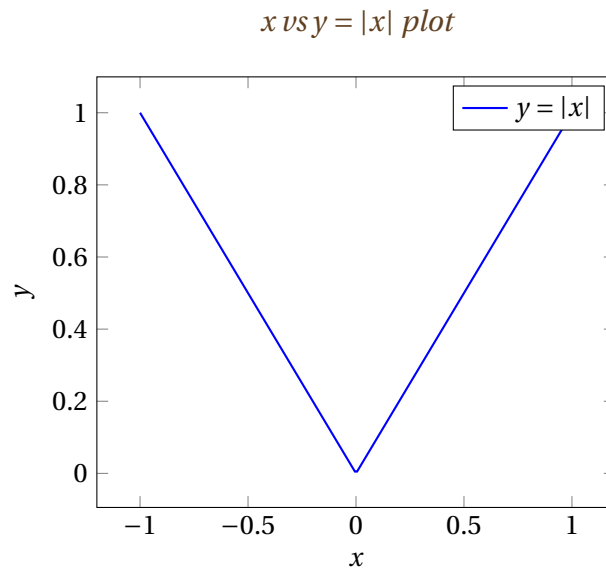
- Axis line style:** The appearance of the axis lines can be customized using the `axis line style` option. Some settings are “`axis line style={thick}`, `axis line style={dashed}`, `axis line style={->}`”.
- Equal Axis Scaling:** Sometimes it is important that one unit on the *x* axis corresponds to the same physical length as one unit on the *y*-axis. This can be enforced using `axis equal`.
- Title:** We can give a title to our plot using `title={My Plot}` command. We can change style the title of our plot using `title style={style options}` command.

Here is a minimal example which explores how to add title and style it.

```

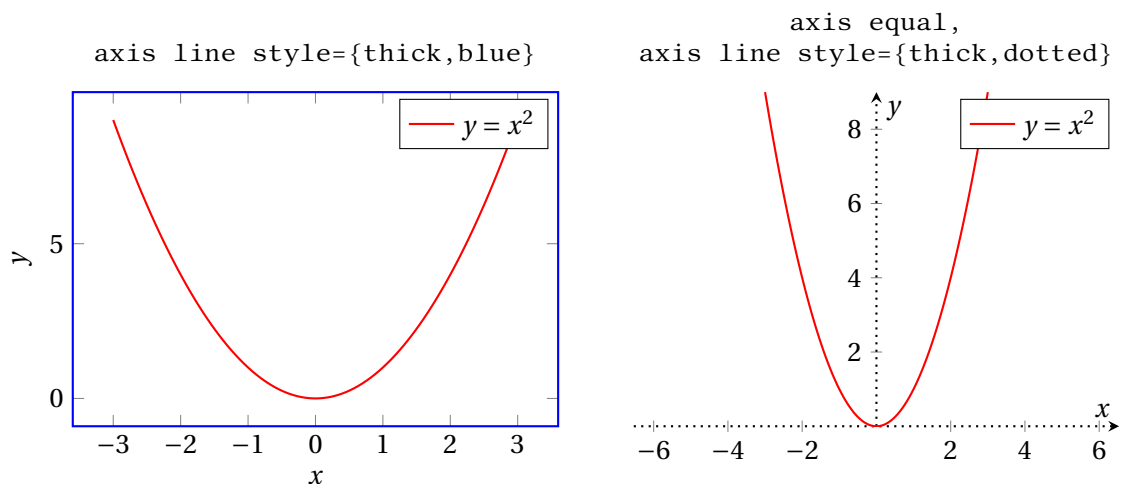
1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[xlabel=$x$,ylabel=$y$,title={$x\,,\,\text{vs}\,,\,y=\abs{x}$ plot},
4 title style={color=brown!50!black,font=\large\itshape,yshift=5pt}]
5 \addplot[domain=-1:1,samples=200,blue,thick]{abs(x)};
6 \addlegendentry{$y=\abs{x}$}
7 \end{axis}
8 \end{tikzpicture}
9 \end{center}

```



- (a) The line `title={$x\,,\,\text{vs}\,,\,y=\abs{x}$ plot}` adds the title to the plot.
- (b) `title style={color=brown!50!black,font=\large\itshape,yshift=5pt}` does three things. First it changes the color, then the font which is made large and italic and we shift the title along y axis by 5 pt.

These options help control the visual style and readability of plots created using PGFPlots. The below figure explores “Axis lines style” options and “Equal axis scaling” options.



Note that I haven't provided any codes for generating these plots. We will come to this when we will deal with subplots.

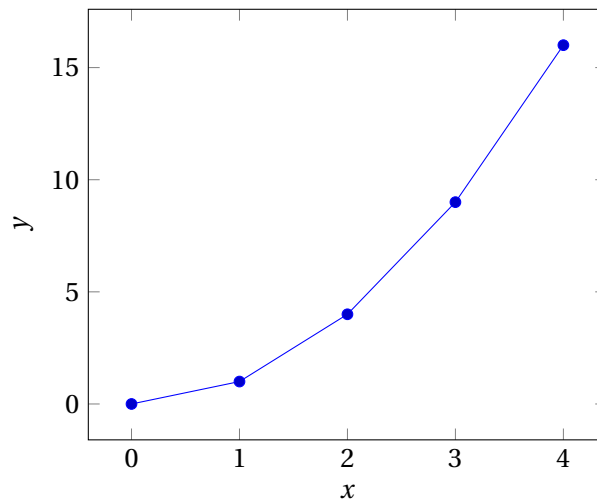
1.13 Plotting using Data Coordinates and External Tables

Plots are not limited to mathematical functions. We can also plot experimental data, simulation results, or tabulated values. This can be done in two main ways: Using data coordinates directly in the document, Using external data tables.

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[xlabel=$x$,ylabel=$y$]
4 \addplot coordinates {(0,0)(1,1)(2,4)(3,9)(4,16)};
5 \end{axis}
6 \end{tikzpicture}
7 \end{center}

```



Each pair (x, y) represents one data point in the plot. This method is convenient when the dataset is small. When working with large datasets, it is better to store the data in an external file and import it into the plot. Suppose we have a file “data.dat” containing the data sets (shown in Table. 1.1):

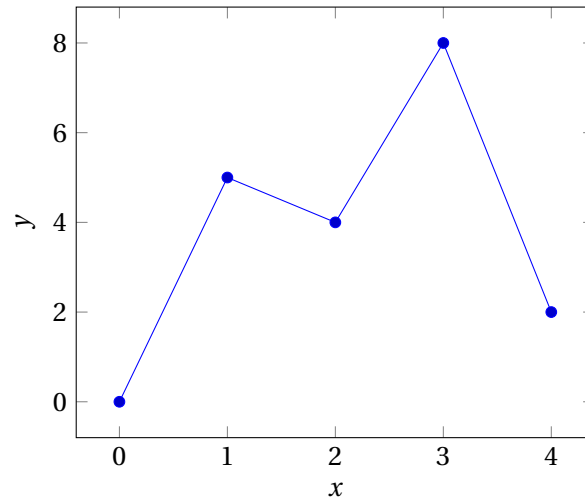
x	y
0	0
1	5
2	4
3	8
4	2

Table 1.1: Dataset containing two variable x and y .

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[xlabel=$x$,ylabel=$y$]
4 \addplot table {data.dat};
5 \end{axis}
6 \end{tikzpicture}
7 \end{center}

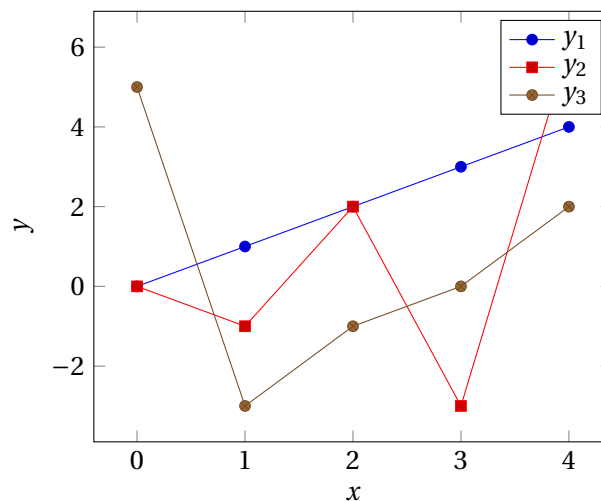
```



If the table contains multiple columns, we can specify which columns to use. For example using command `\addplot table [x=x,y=y] {data.dat}`; we can achieve this. Here `x=x` means use column `x` for the horizontal axis and `y=y` means use column `y` for the vertical axis. Suppose we have more than two columns in the dataset called `data1.dat` as shown in Table 1.2.

x	y_1	y_2	y_3
0	0	0	5
1	1	-1	-3
2	2	2	-1
3	3	-3	0
4	4	6	2

Table 1.2: Dataset containing four variables x, y_1, y_2 and y_3 .



Sometimes the table only contains numbers. Then we refer to columns by index:

1. `\addplot table [x index=0,y index=1] {data1.dat}`;

2. `\addplot table [x index=0,y index=2] {data1.dat}`;

Here `x index=0` → first column, `y index=1` → second column, `y index=2` → third column.

1.14 Parametric Plots

Parametric plot is used when both coordinates are expressed as functions of a third variable (usually denoted by t). Instead of writing $y = f(x)$, we describe the curve as

$$x = x(t), \quad y = y(t) \quad (1.1)$$

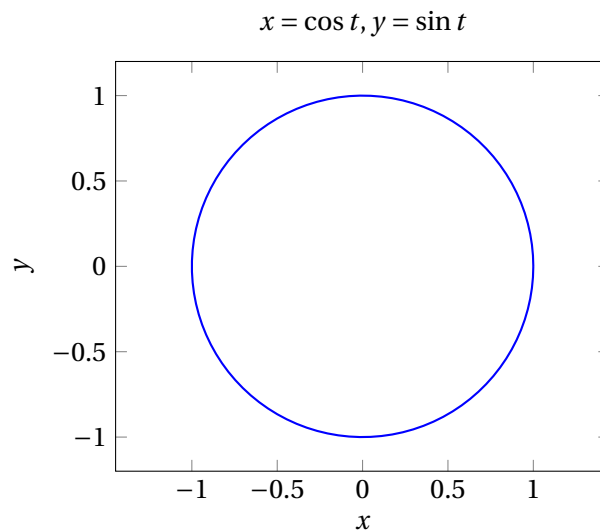
Such plots are very common in physics and mathematics. For example circular motion, trajectories, Lissajous curves, spirals. Suppose we want to draw a circle of radius 1 can be written as :

$$x = \cos t, \quad y = \sin t \quad (1.2)$$

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[xlabel=$x$,ylabel=$y$,axis equal,title={$x=\cos\{t\},y=\sin\{t\}$},height=7cm]
4 \addplot[domain=0:360,samples=100,blue,thick]({cos(x)},{sin(x)});
5 \end{axis}
6 \end{tikzpicture}
7 \end{center}

```



`\addplot[domain=0:360,samples=100,blue,thick]({cos(x)},{sin(x)})`; is the main line which generates this parametric plot. Here the variable x acts as the parameter t . For parametric curves, it is often helpful to use `axis equal` which ensures that the scaling of the x and y axes is the same. This prevents shapes like circles from appearing distorted.

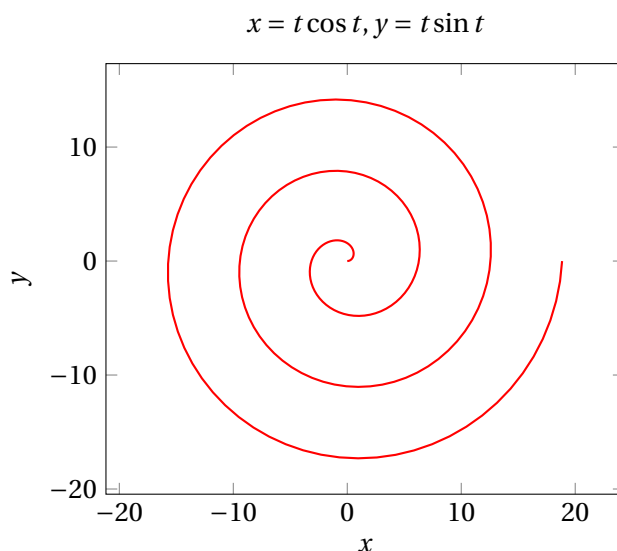
A spiral can be written as

$$x(t) = t \cos t, \quad y(t) = t \sin t \quad (1.3)$$

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[xlabel=$x$,ylabel=$y$,axis equal,title={$x=t\cos\{t\},y=t\sin\{t\}$}]
4 \addplot[domain=0:6*pi,samples=200, red, thick]({x*cos(deg(x))},{x*sin(deg(x))});
5 \end{axis}
6 \end{tikzpicture}
7 \end{center}

```



The command `domain=0:6*pi` makes the range of $t = [0, 6\pi]$.

1.15 Polar Plots

In PGFPlots, polar plots are used when a curve is described in polar coordinates instead of Cartesian coordinates. In polar form, a curve is written as

$$r = f(\theta) \quad (1.4)$$

where r is the distance from the origin and θ is the angle. Polar plots are commonly used in physics for radiation patterns, orbital shapes, spiral curves.

To create a polar plot, we use the option `polaraxis` instead of the usual `axis` environment. In order to use it you have to add this after loading “PGFPlots”.

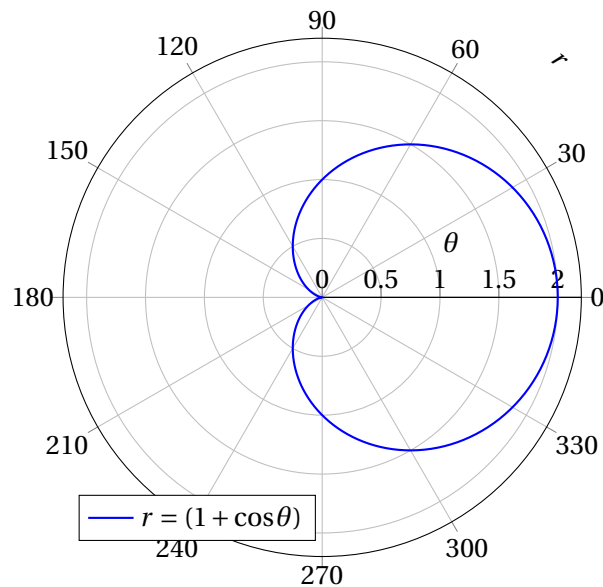
```
1 \usepgfplotslibrary{polar}
```

A very popular polar curve is the cardioid,

$$r = a(1 + \cos \theta) \quad (1.5)$$

which often appears in optics and wave interference patterns. Note that angles in polar plots are measured in degrees by default.

```
1 \begin{center}
2 \begin{tikzpicture}
3 \begin{polaraxis}[xlabel={\textcolor{red}{$r$}}, ylabel={\textcolor{blue}{$\theta$}}, legend pos = south west]
4 \addplot[domain=0:360,samples=200,blue,thick]{1 + cos(x)};
5 \addlegendentry{\textcolor{red}{$r$}=(1+\textcolor{blue}{cos}\textcolor{red}{$\theta$})\textcolor{blue}{$$}}
6 \end{polaraxis}
7 \end{tikzpicture}
8 \end{center}
```



1.16 Make Subplots

In many scientific documents we often need to display multiple plots together for comparison. In PGFPlots, this can be done using the group plot environment, which allows us to arrange several plots in a grid layout. This feature is provided by the `groupplots` library. To use grouped plots, add the following in the preamble:

```
1 \usepgfplotslibrary{groupplots}
```

The structure for multiple plots is

```
1 \begin{center}
2 \begin{tikzpicture}
3 \begin{groupplot}[group style={group size=<columns> by <rows>}]
4
5 \nextgroupplot<first plot>
6
7 \nextgroupplot<second plot>
8 .....
9 \end{groupplot}
10 \end{tikzpicture}
11 \end{center}
```

Consider you want a total of 4 plots, total 2 rows and in each row 2 columns. In the first row you want to plot $y = x$ and $y = x^2$, two different columns. Again in the next row plot $y = \sin x$ and $y = \cos x$. As shown before the structure can be taken as follows :

Row/Col	Col 1	Col 2
Row 1	first plot $y = x$	second plot $y = x^2$
Row 2	third plot $y = \sin x$	fourth plot $y = \cos x$

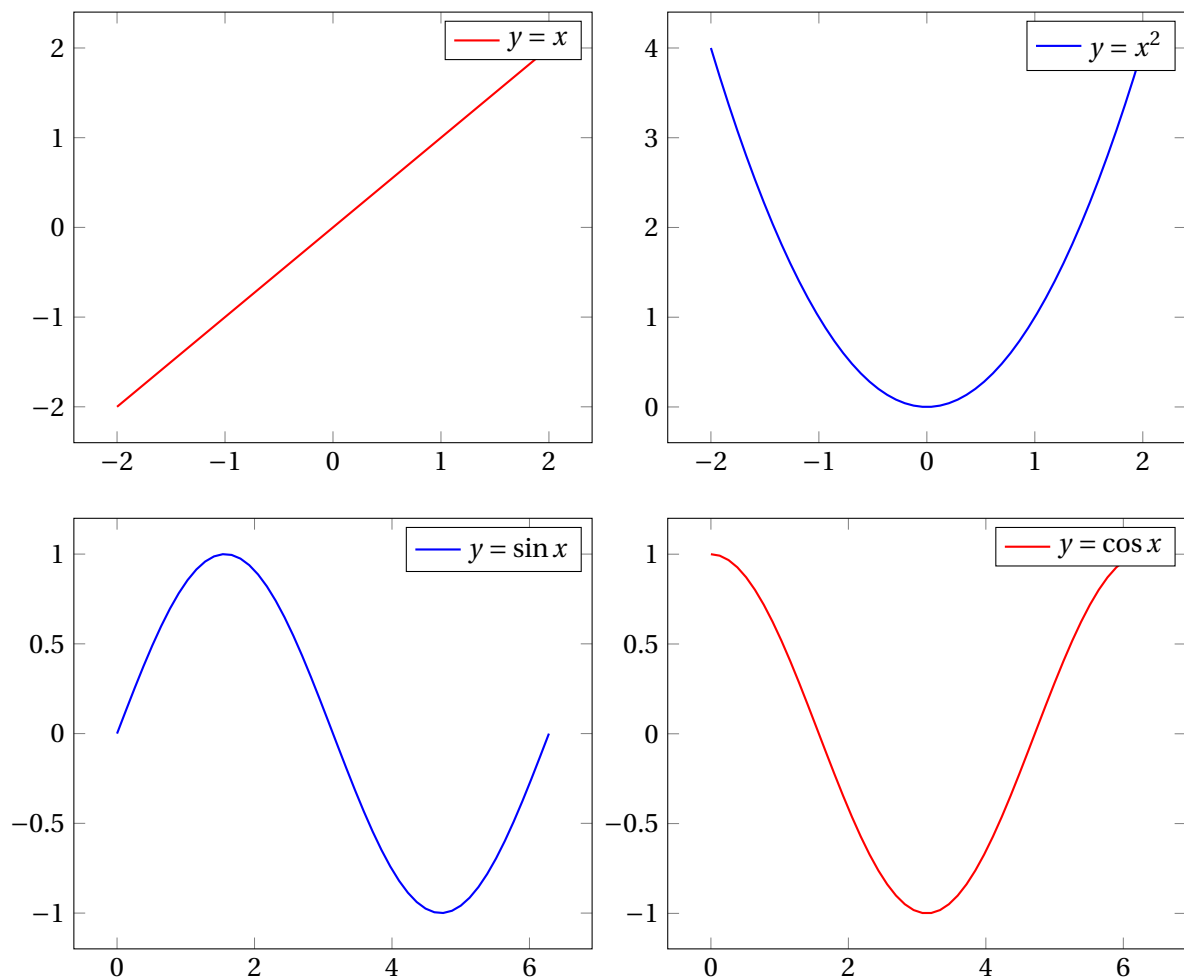
Table 1.3: Subplots diagram

`\nextgroupplot` is the command to plot different subplots.

```

1  \begin{center}
2  \begin{tikzpicture}
3  \begin{groupplot}[
4  group style={group size=2 by 2}]
5
6  \nextgroupplot
7  \addplot[domain=-2:2,samples=50,red,thick] {x};
8  \addlegendentry{$y=x$}
9  \nextgroupplot
10 \addplot[domain=-2:2,samples=50,blue,thick] {x^2};
11 \addlegendentry{$y=x^2$}
12
13 \nextgroupplot
14 \addplot[domain=0:2*pi,samples=50,blue,thick] {sin(deg(x))};
15 \addlegendentry{$y=\sin{x}$}
16 \nextgroupplot
17 \addplot[domain=0:2*pi,samples=50,red,thick] {cos(deg(x))};
18 \addlegendentry{$y=\cos{x}$}
19 \end{groupplot}
20 \end{tikzpicture}
21 \end{center}

```



Now we will discuss how we generated the figures in Sec. 1.12. The first figure was generated via :

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{groupplot}[group style={group size=3 by 1},width=6cm,height=6cm]
4 \nextgroupplot[legend pos=north west,title=\texttt{legend pos=north west}]
5 \addplot[domain=-1:1,samples=100,red,thick]{x};
6 \addlegendentry{$y=x$}
7 \nextgroupplot[axis lines=middle,title=\texttt{axis lines=middle}]
8 \addplot[domain=-1:1,samples=100,red,thick]{-x};
9 \addlegendentry{$y=-x$}
10 \nextgroupplot[axis lines=left,legend pos=south east,
11 title=\shortstack{\texttt{axis lines=left},}\texttt{legend pos=south east}}]
12 \addplot[domain=-1:1,samples=100,red,thick]{x};
13 \addlegendentry{$y=x$}
14 \end{groupplot}
15 \end{tikzpicture}
16 \end{center}

```

Let's analyze :

1. The line `[group style={group size=3 by 1},width=6cm,height=6cm]` creates 2 columns and 1 row, each having width of 6 cm and height 6 cm.
2. By changing `\nextgroupplot[options]` you can customize each subplot.
3. Each `\nextgroupplot[options]` behaves like the axis environment.
4. `group style={horizontal sep=<length>,vertical sep=<length>}` can be used to adjust the separations between the subplots.

Let's do one more example to give you a better idea. Consider total 4 plots with 2 rows and 2 columns. We will be plotting the same functions as we did at first, but instead of all having same width and height, we will give different height and widths. We accomplish it by this way :

```

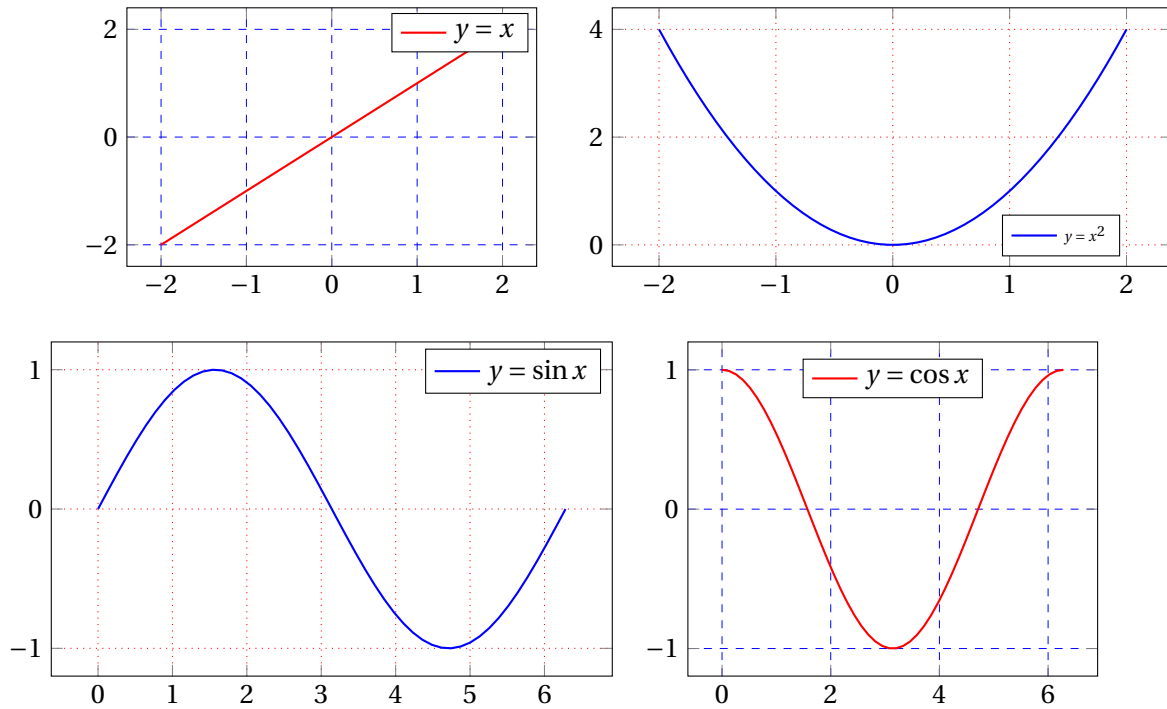
1 \begin{center}
2 \begin{tikzpicture}
3 \begin{groupplot}[
4 group style={group size=2 by 2},width=8cm,height=5cm]
5 \nextgroupplot[width=7cm,grid=major,major grid style={dashed,very thin,color=blue}]
6 \addplot[domain=-2:2,samples=50,red,thick]{x};
7 \addlegendentry{$y=x$}
8 \nextgroupplot[width=9cm,grid=major,major grid style={dotted,thin,color=red},
9 legend style={font=\tiny,at={(0.8,0.2)},anchor=north}]
10 \addplot[domain=-2:2,samples=50,blue,thick]{x^2};
11 \addlegendentry{$y=x^2$}
12 \nextgroupplot[width=9cm,height=6cm,grid=major,
13 major grid style={dotted,thin,color=red}]
14 \addplot[domain=0:2*pi,samples=50,blue,thick]{sin(deg(x))};
15 \addlegendentry{$y=\sin{x}$}

```

```

16 \nextgroupplot[width=7cm,height=6cm,grid=major,
17 major grid style={dashed,very thin,color=blue},
18 legend style={at={(0.5,0.95)},anchor=north}]
19 \addplot[domain=0:2*pi,samples=50,red,thick] {cos(deg(x))};
20 \addlegendentry{$y=\cos{x}$}
21 \end{groupplot}
22 \end{tikzpicture}
23 \end{center}

```



By `\nextgroupplot[width=7cm]`, `\nextgroupplot[width=9cm,height=6cm]`, we change the size of each subplots.

1.17 Declare variables and Create Functions

It is often useful to define variables and reusable mathematical functions. This makes the code cleaner and allows complex expressions to be reused in multiple plots. In PGFPlots (built on TikZ), there are a few different ways to define variables, macros, and functions using `\pgfmath`. It is important to distinguish between:

- ✓ `\pgfmathsetmacro` → defines a macro containing a numerical value.
- ✓ `\pgfmathdeclarefunction` → defines a reusable mathematical function.

1. `\pgfmathsetmacro{\macro}{expression}` is the command to define a variable. For example, define a variable a and sets the value to 2 can be done using `\pgfmathsetmacro{\a}{2}`.
2. `\pgfmathdeclarefunction{name}{number of arguments}{definition}` is the command to define a function. Say we want to define a function, say $f(x) = x^2 + 2x + 1$. This can be achieved by following:

```

1 \pgfmathdeclarefunction{f}{1}{%
2 \pgfmathparse{#1^2 + 2*#1 + 1}}

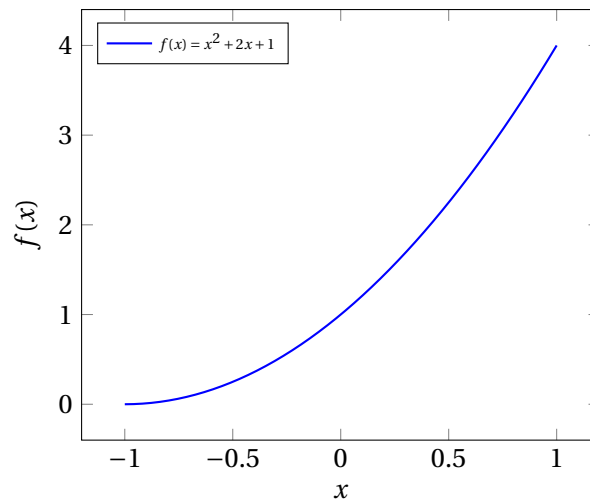
```

{1} means that only one variable and #1 refers to that variable in the expression. Here is the simplest code to show this :

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[xlabel=$x$,ylabel=$f(x)$,legend pos=north west,legend style={font=\tiny}]
4 % declare function $f(x)=x^2+2x+1$
5 \pgfmathdeclarefunction{f}{1}{%
6 \pgfmathparse{#1^2 + 2*#1 + 1}}
7 \addplot[domain=-1:1,samples=50,blue,thick]{f(x)};
8 \addlegendentry{$f(x)=x^2+2x+1$}
9 \end{axis}
10 \end{tikzpicture}
11 \end{center}

```



We now move on to some examples.

1.17.1 Central Field Potential

Consider the 2D motion of particle of mass m in a central potential $V(r)$. The effective potential is given by :

$$V_{\text{eff}}(r) = V(r) + \frac{L^2}{2mr^2} \quad (1.6)$$

The 2nd term is known as “centrifugal term”. For a particle moving in a gravitational field this form is given as,

$$V_{\text{eff}}(r) = \underbrace{-\frac{GMm}{r}}_{V(r)} + \underbrace{\frac{L^2}{2mr^2}}_{V_c(r)} \quad (1.7)$$

To analyze how this potential looks like, we will plot V_r , $V_c(r)$ and $V_{\text{eff}}(r)$ all together.

```

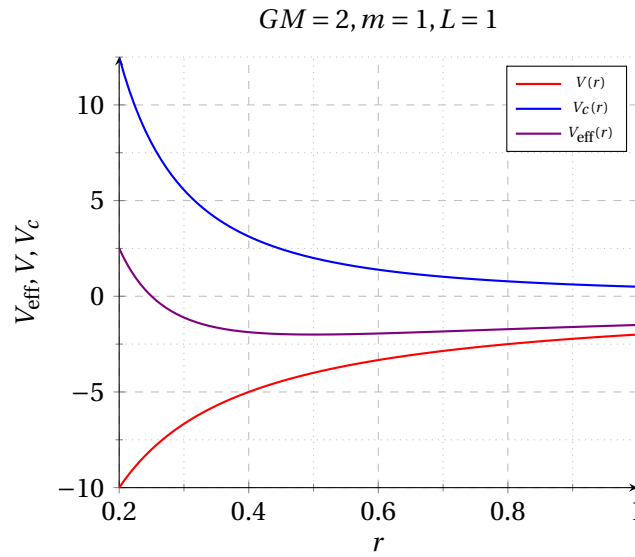
1  \begin{center}
2  \begin{tikzpicture}
3
4  % define parameters GM,m,L
5  \pgfmathsetmacro{\GM}{2} % GM
6  \pgfmathsetmacro{\m}{1}  % m
7  \pgfmathsetmacro{\L}{1}  % L
8
9  \begin{axis}[xlabel=$r$,ylabel={$V_{\mathrm{eff}}$},V,V_c$,
10 grid=both,major grid style={dashed},
11 minor grid style={dotted},minor tick num=1,
12 axis lines=left,legend style={font=\tiny},
13 domain=0.2:1,samples=100,title={$GM=\GM,m=\m,L=\L$}]
14 % define potentials
15 %  $V(r) = -\frac{GMm}{r}$ 
16 \pgfmathdeclarefunction{vr}{1}{%
17 \pgfmathparse{-(\GM*\m)/\#1}}
18
19 %  $V_c(r) = \frac{L^2}{2mr^2}$ 
20 \pgfmathdeclarefunction{vcr}{1}{%
21 \pgfmathparse{(\L^2)/(2*\m*\#1*\#1)}}
22
23 %  $V_{\mathrm{eff}}(r)$ 
24 \pgfmathdeclarefunction{veffr}{1}{%
25 \pgfmathparse{vr(\#1)+vcr(\#1)}}
26
27 % plot them
28 %  $V(r)$ 
29 \addplot[thick,red]{vr(x)};
30 \addlegendentry{$V(r)$}
31
32 %  $V_c(r)$ 
33 \addplot[thick,blue]{vcr(x)};
34 \addlegendentry{$V_c(r)$}
35
36 %  $V_{\mathrm{eff}}(r)$ 
37 \addplot[thick,red!50!blue]{veffr(x)};
38 \addlegendentry{$V_{\mathrm{eff}}(r)$}
39 \end{axis}
40 \end{tikzpicture}
41 \end{center}

```

Let's analyze :

- (a) `legend style={font=\tiny}` sets the “tiny” fontsize.
- (b) We have added this `domain=0.2:1,samples=100` command inside the axis environment. This setting will be applied to each `\addplot[options]` command.

(c) `title=\{GM=\backslash GM,m=\backslash m,L=\backslash L\}` takes the values from the definitions automatically.



1.17.2 Plotting a Gaussian

In many areas of physics we need to plot Gaussian functions. Gaussian function with mean μ and standard deviation σ is given by :

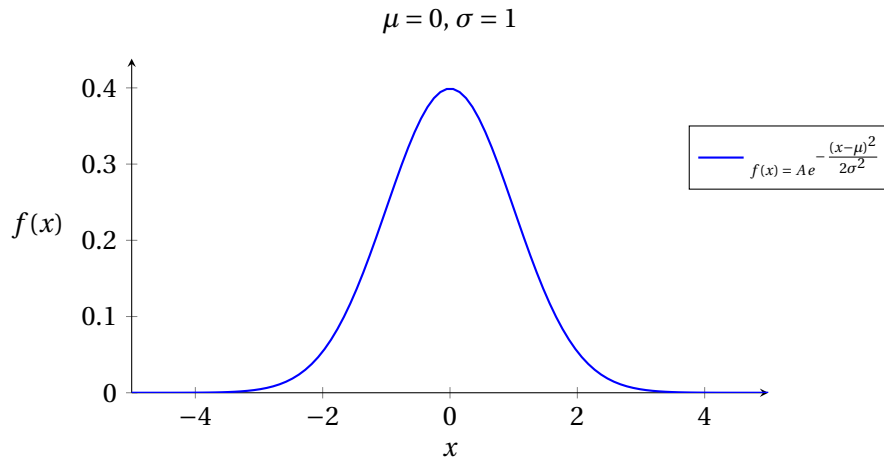
$$f(x) = A e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (1.8)$$

If normalized $A = \frac{1}{\sqrt{2\pi\sigma^2}}$. Lets plot this gaussian having $\mu = 0$ and $\sigma = 1$.

```

1  \begin{center}
2  \begin{tikzpicture}
3  \pgfmathsetmacro{\mean}{0} % define mu
4  \pgfmathsetmacro{\std}{1} % define sigma
5  \begin{axis}[xlabel=$x$,ylabel=$f(x)$,title=({\mu=\mean,\,\sigma=\std}),height=6cm,
6  width=10cm,axis lines=left,ylabel style={rotate=270},legend style={at={(1.2,0.8)}},
7  font=\tiny,enlarge y limits={upper=0.1}]
8
9  % get the const A
10 \pgfmathsetmacro{\A}{1/(sqrt(2*pi*\std^2))}
11 % define the gaussian
12 \pgfmathdeclarefunction{gauss}{1}{%
13 \pgfmathparse{\A*exp(-((#1-\mean)^2)/(2*\std^2))}}
14
15 % plot the gaussian
16 \addplot[domain=-5:5,samples=100,blue,thick] {gauss(x)};
17 \addlegendentry{$f(x)=A\cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$}
18 \end{axis}
19 \end{tikzpicture}
20 \end{center}

```



1.18 Roses are Red

In mathematics, a “rose or rhodonea” curve is a sinusoid specified by either the cosine or sine functions with no phase angle that is plotted in polar coordinates. Rose curves or "rhodonea" were named by the Italian mathematician who studied them, Guido Grandi, between the years 1723 and 1728. A rose is the set of points in polar coordinates specified by the polar equation

$$r = a f(k\theta) \quad (1.9)$$

Or in cartesian coordinates, the same equation given as :

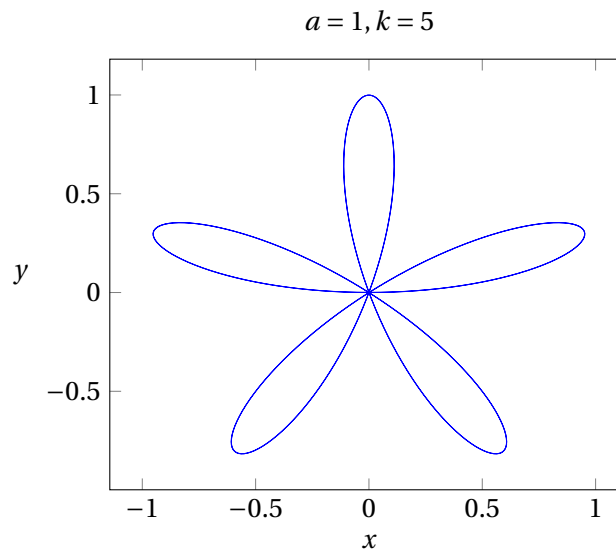
$$x = r \cos \theta = a f(k\theta) \cos \theta \quad (1.10a)$$

$$y = r \sin \theta = a f(k\theta) \sin \theta \quad (1.10b)$$

The function f is either sin or cos. Now if

1. $k = \text{odd}$, number of petals = k .
2. $k = \text{even}$, number of petals = $2k$.

When plotted over the θ range $[0, 2\pi]$ they produce full curve. Lets plot it for $k = 5$, with $f = \sin$.



```

1 \begin{center}
2 \begin{tikzpicture}
3 \pgfmathsetmacro{\a}{1} % define a
4 \pgfmathsetmacro{\k}{5} % define k
5 \begin{axis}[xlabel=$x$,ylabel=$y$,title={$a=\a,k=\k$},ylabel style={rotate=270}]
6
7 % define x = xvar
8 \pgfmathdeclarefunction{xvar}{1}{%
9 \pgfmathparse{\a*sin(\k*#1)*cos(#1)}}
10 % define y = yvar
11 \pgfmathdeclarefunction{yvar}{1}{%
12 \pgfmathparse{\a*sin(\k*#1)*sin(#1)}}
13
14 % plot the rose
15 \addplot[domain=0:360,samples=300,blue] ({xvar(x)},{yvar(x)});
16 \end{axis}
17 \end{tikzpicture}
18 \end{center}

```

1.19 An Home-made example

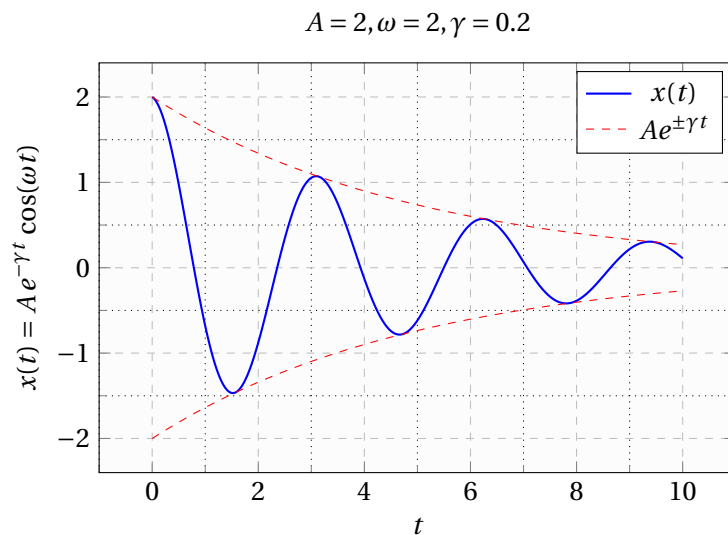
A damped harmonic oscillator is a system where an oscillating motion gradually decreases in amplitude over time because of a damping force (like friction or air resistance). The position as a function of time is given as :

$$x(t) = A e^{-\gamma t} \cos(\omega t + \phi) \quad (1.11)$$

Here A, ϕ to be determined from the initial conditions and $\gamma = \frac{b}{2m}, \omega = \sqrt{\omega_0^2 - \gamma^2}, \omega_0 = \sqrt{\frac{k}{m}}$. b is the damping coefficient, m is the mass and k is the spring constant. Take initial conditions as $x(t=0) = A$ and $\dot{x}(t=0) = 0$, which gives :

$$x(t) = A e^{-\gamma t} \cos(\omega t) \quad (1.12)$$

Here we have plotted $x(t)$ for t in range $[0, 10]$ sec.



The code that was used to produce is the following :

```

1 \begin{center}
2 \begin{tikzpicture}
3
4 % define parameters
5 \pgfmathsetmacro{\A}{2} % amplitude A
6 \pgfmathsetmacro{\g}{0.2} % damping factor gamma
7 \pgfmathsetmacro{\w}{2} % angular frequency omega
8
9 % damped oscillator solution  $x(t) = A e^{-\gamma t} \cos(\omega t)$ 
10 \pgfmathdeclarefunction{xt}{1}{%
11     \pgfmathparse{\A*cos(\w*#1 r)*exp(-\g*#1)}}
12
13 % envelopes  $A e^{\pm \gamma t}$ 
14 \pgfmathdeclarefunction{envelopeplus}{1}{%
15     \pgfmathparse{\A*exp(-\g*#1)}}
16
17 \pgfmathdeclarefunction{envelopeminus}{1}{%
18     \pgfmathparse{-\A*exp(-\g*#1)}}
19
20 % draw plot
21 \begin{axis}[title={ $A=\A, \omega=\w, \gamma=\g$ }, domain=0:10, samples=500,
22 grid=both,axis background/.style={fill=gray!2},major grid style={dashed},
23 minor tick num = 1,minor grid style={dotted,black},height=7cm,width=10cm,
24 xlabel= $t$ ,ylabel={ $x(t)=A e^{-\gamma t} \cos(\omega t)$ }]
25
26 % draw  $x(t)$ 
27 \addplot[blue, thick] {xt(\x)};
28
29 % draw envelopes
30 \addplot[red, dashed] {envelopeplus(\x)};
31 \addplot[red, dashed] {envelopeminus(\x)};
32
33 % add legends
34 \legend{{ $x(t)$ },  $A e^{\pm \gamma t}$ }
35
36 \end{axis}
37 \end{tikzpicture}
38 \end{center}

```

`axis background/.style={fill=gray!2}`, is the command that fills the background with 2% gray color.

1.20 Advanced Problems

Here we will discuss some more advanced plots, that will help you to master “PGFPlots”. This will help you to develop more about the functions, axis styles and much more.

1.20.1 Butterfly Plot

The butterfly curve is a famous transcendental curve discovered by Temple H. Fay (1989). It has a shape resembling a butterfly when plotted in polar coordinates. One common form of the butterfly curve is :

$$x = \sin t \left[e^{\cos t} - 2 \cos 4t - \sin^5 \left(\frac{t}{12} \right) \right] \quad (1.13a)$$

$$y = \cos t \left[e^{\cos t} - 2 \cos 4t - \sin^5 \left(\frac{t}{12} \right) \right] \quad (1.13b)$$

Or in polar form we obtain :

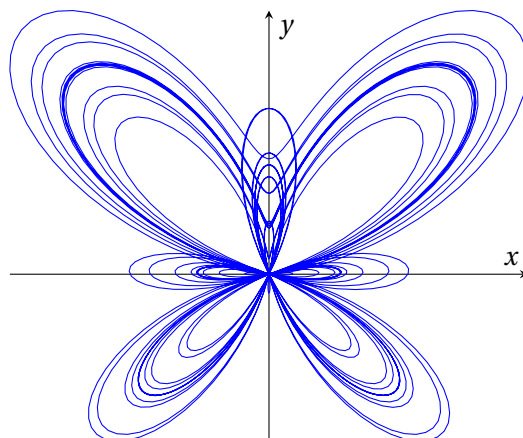
$$r = e^{\sin \theta} - 2 \cos(4\theta) + \sin^5 \left(\frac{2\theta - \pi}{24} \right) \quad (1.14)$$

Where we have defined $x = r \cos \theta$, $y = r \sin \theta$, with $\theta = \frac{\pi}{2} - t$. The curve is usually plotted for $0 \leq t \leq 24\pi$ to obtain the full butterfly structure.

```

1  \begin{center}
2  \begin{tikzpicture}
3  \begin{axis}[axis lines = middle,xtick=\empty,ytick=\empty,
4  xlabel=$x$,ylabel=$y$]
5
6  % define polar function rvar = r(\theta)
7  \pgfmathdeclarefunction{rvar}{1}{%
8  \pgfmathparse{exp(sin(deg(#1)))-2*cos(deg(4*#1))+(sin(deg((2*#1-pi)/24)))^5}}
9
10 % define xvar=x(t) and yvar=y(t)
11 \pgfmathdeclarefunction{xvar}{1}{%
12 \pgfmathparse{sin(deg(#1))*rvar((pi/2)-#1)}}
13 \pgfmathdeclarefunction{yvar}{1}{%
14 \pgfmathparse{cos(deg(#1))*rvar((pi/2)-#1)}}
15
16 % plot the butterfly
17 \addplot[domain=0:24*pi,samples=2000,blue] ({xvar(x)},{yvar(x)});
18 \end{axis}
19 \end{tikzpicture}
20 \end{center}

```



1.20.2 Simple Harmonic Oscillator

Consider “simple harmonic oscillator” of mass m and natural frequency ω . We have the position and momentum as :

$$q(t) = q_0 \sin \omega t \quad (1.15a)$$

$$p(t) = m\dot{x} = p_0 \cos \omega t, \quad p_0 = m\omega q_0 \quad (1.15b)$$

Now in classical mechanics, we often do “canonical transformation” which simplifies equation of motion. In canonical coordinates $(q, p) \rightarrow (Q, P)$ which are related by :

$$q = \sqrt{\frac{2P}{m\omega}} \sin Q \quad (1.16a)$$

$$p = \sqrt{2Pm\omega} \cos Q \quad (1.16b)$$

Which gives you the solution as :

$$P(t) = P_0 = \text{const} \quad (1.17a)$$

$$Q(t) = \omega t \quad (1.17b)$$

We can easily get $P_0 = \frac{m\omega q_0^2}{2}$. Now what we want to do is make 3 rows, 2 columns each. Plot it this way :

	Old	New
Position	t vs $q(t)$	t vs $Q(t)$
Momentum	t vs $p(t)$	t vs $P(t)$
Phase Space	$q(t)$ vs $p(t)$	$Q(t)$ vs $P(t)$

Table 1.4: SHO diagram

Here we have taken $q_0 = \omega = 1$ and $m = 2$.

```

1 \begin{center}
2 \begin{tikzpicture}
3
4 % define parameters
5 \pgfmathsetmacro{\qnot}{1} % amplitude q_0
6 \pgfmathsetmacro{\w}{1} % angular frequency omega
7 \pgfmathsetmacro{\m}{2} % unit mass
8 \pgfmathsetmacro{\pnot}{\m*\w*\qnot} % p_0=m*omega*q_0
9 \pgfmathsetmacro{\Pnot}{0.5*\m*\w*\qnot*\qnot} % P_0=\frac{m*omega*q_0^2}{2}
10
11 % define old variables
12 \pgfmathdeclarefunction{qvar}{1}{%
13 \pgfmathparse{\qnot*sin(deg(\w*#1))}}
14
15 \pgfmathdeclarefunction{pvar}{1}{%
16 \pgfmathparse{\pnot*cos(deg(\w*#1))}}
```

```

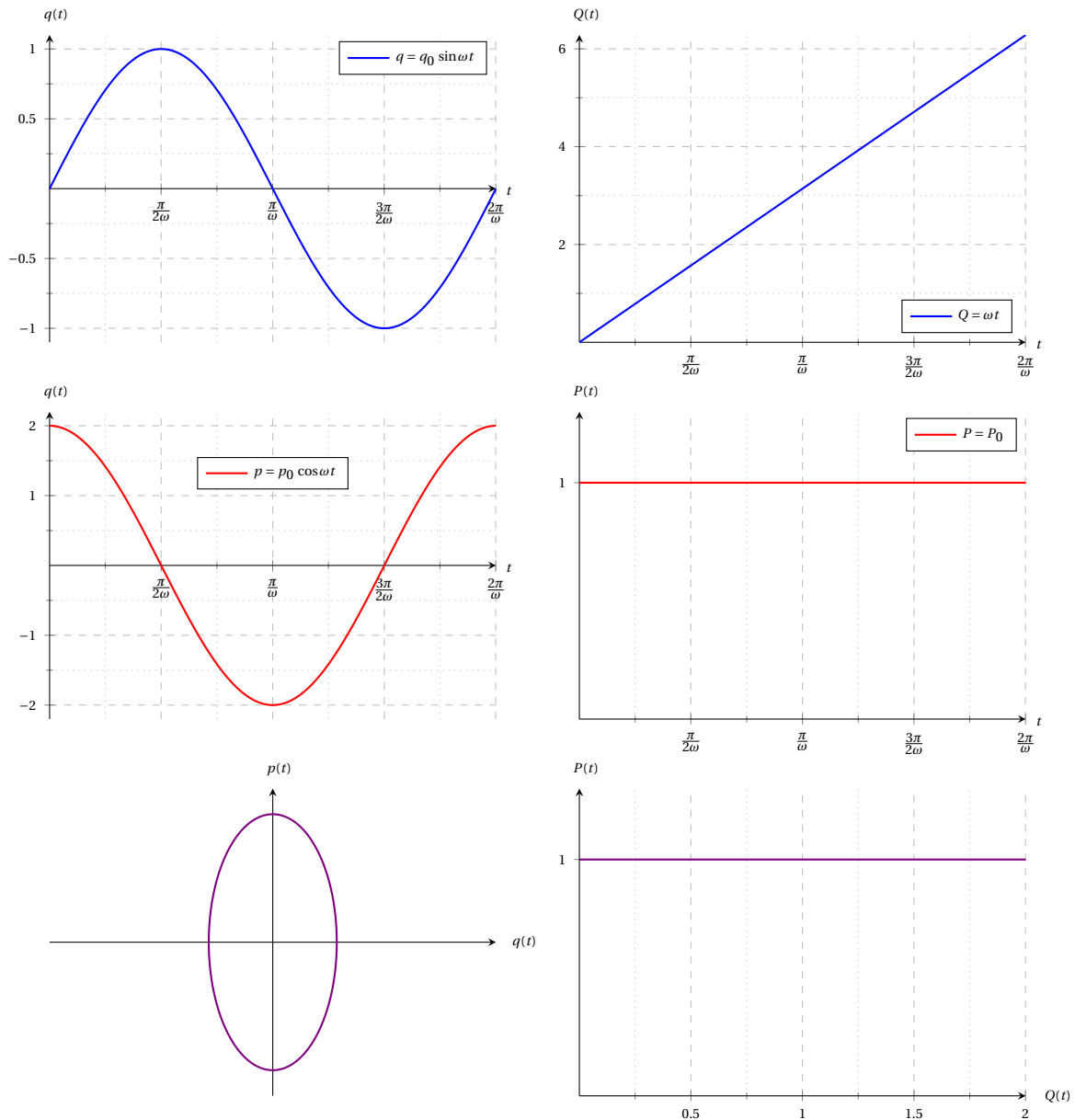
17
18 % define new variables
19 \pgfmathdeclarefunction{Qvar}{1}{%
20 \pgfmathparse{\w*#1}}
21
22 \pgfmathdeclarefunction{Pvar}{1}{%
23 \pgfmathparse{\Pnot}}
24
25 \begin{groupplot}[group style={group size=2 by 3,horizontal sep= 12mm},
26 height=6cm,width=8cm,domain=0:2*pi,samples=200,
27 axis lines=center,xlabel=$t$,
28 xlabel style={font=\tiny,xshift=10pt,yshift=-6pt},
29 ylabel style={font=\tiny,xshift=-6pt,yshift=14pt},
30 legend style={font=\tiny},xticklabel style={font=\tiny},
31 yticklabel style={font=\tiny},xtick={0,1.57,3.14,4.71,6.28},
32 xticklabels={$0$, $\tfrac{\pi}{2\omega}$, $\tfrac{\pi}{\omega}$,
33 $\tfrac{3\pi}{2\omega}$, $\tfrac{2\pi}{\omega}$},
34 grid=both,major grid style={dashed},minor grid style={dotted},minor tick num=1]
35
36 % q vs t
37 \nextgroupplot[ylabel=$q(t)$,ymin=-1.1*\qnot,ymax=1.1*\qnot]
38 \addplot[blue,thick]{qvar(x)};
39 \addlegendentry{$q=q_0\backslash,\sin{\omega t}$}
40
41 % Q vs t
42 \nextgroupplot[ylabel=$Q(t)$,legend pos=south east]
43 \addplot[blue,thick]{Qvar(x)};
44 \addlegendentry{$Q=\omega t$}
45
46 % p vs t
47 \nextgroupplot[ylabel=$p(t)$,legend style={at={(0.5,0.8)},anchor=center},
48 ylabel=$q(t)$,ymin=-1.1*\pnot,ymax=1.1*\pnot]
49 \addplot[red,thick]{pvar(x)};
50 \addlegendentry{$p=p_0\backslash,\cos{\omega t}$}
51
52 % P vs t
53 \nextgroupplot[ylabel=$P(t)$,ymin=0,ymax=1.3*\Pnot,ytick={\Pnot}]
54 \addplot[red,thick]{Pvar(x)};
55 \addlegendentry{$P=P_0$}
56
57 % p vs q
58 \nextgroupplot[axis equal,xlabel=$q(t)$,
59 xlabel style={font=\tiny,xshift=10pt,yshift=0pt},
60 ylabel=$p(t)$,ymin=-1.2*\pnot,ymax=1.2*\pnot,
61 xtick=\empty,ytick=\empty]
62 \addplot[red!50!blue,thick]({qvar(x)},{pvar(x)});
63
64 % P vs Q
65 \nextgroupplot[xlabel=$Q(t)$,xlabel style={font=\tiny,xshift=12pt,yshift=0pt},

```

```

66 ylabel=$P(t)$, ymin=0, ymax=1.3*\Pnot, ytick={\Pnot},
67 xticklabels={$0$, $0.5$, $1$, $1.5$, $2$}]
68 \addplot[red!50!blue, thick] ({Qvar(x)}, {Pvar(x)});
69
70 \end{groupplot}
71 \end{tikzpicture}
72 \end{center}

```



1.20.3 Two Oscillators

Consider two simple harmonic oscillators, whose positions are given by :

$$x_1(t) = A_1 \sin(\omega_1 t) \quad (1.18a)$$

$$x_2(t) = A_2 \sin(\omega_2 t) \quad (1.18b)$$

We would like plot the positions as a function of time. Here is snippet that generates the desired figure :

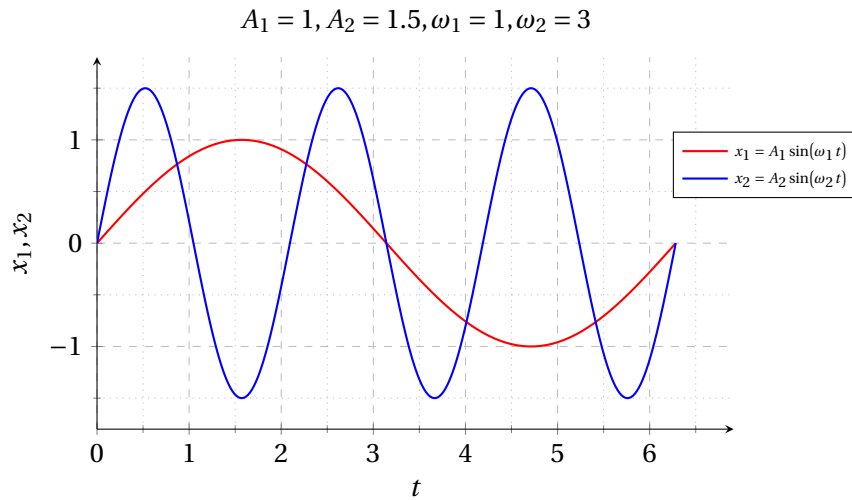
```

1 \begin{center}
2 \begin{tikzpicture}
3
4 % define amplitudes
5 \pgfmathsetmacro{\Aone}{1} % $A_1$
6 \pgfmathsetmacro{\Atwo}{1.5} % $A_2$
7
8 % define frequencies
9 \pgfmathsetmacro{\omegaone}{1} % $\omega_1$
10 \pgfmathsetmacro{\omegatwo}{3} % $\omega_2$
11
12 \begin{axis}[
13 xlabel=$t$,ylabel={$x_1,x_2$},
14 grid=both,major grid style={dashed},
15 minor grid style={dotted},minor tick num=1,
16 axis lines=left,title={$A_1=\textcolor{blue}{\Aone},A_2=\textcolor{blue}{\Atwo},\omega_1=\textcolor{blue}{\omegaone},\omega_2=\textcolor{blue}{\omegatwo}$},
17 legend style={at={(1.2,0.8)},font=\tiny},
18 enlarge x limits={upper=0.2},
19 enlarge y limits=0.1,height=6.5cm,width=10cm]
20
21 % define x_1
22 \pgfmathdeclarefunction{xone}{1}{%
23 \pgfmathparse{\Aone*sin(deg(\omegaone*#1))}}
24
25 % define x_2
26 \pgfmathdeclarefunction{xtwo}{1}{%
27 \pgfmathparse{\Atwo*sin(deg(\omegatwo*#1))}}
28
29 % plot x1 and label it
30 \addplot[domain=0:2*pi,samples=100,red,thick]{xone(x)};
31 \addlegendentry{$x_1=A_1\sin(\textcolor{blue}{\omega_1} t)$}
32
33 % plot x2 and label it
34 \addplot[domain=0:2*pi,samples=300,blue,thick]{xtwo(x)};
35 \addlegendentry{$x_2=A_2\sin(\textcolor{blue}{\omega_2} t)$}
36 \end{axis}
37 \end{tikzpicture}
38 \end{center}

```

Let's point out the meaning of some lines :

- (a) `{legend style={at={(1.2,0.8)},font=\tiny}}` adds legend at the position of (120%,80%) of the whole axis with “tiny” fontsize.
- (b) `enlarge y limits=0.1` adds 10% extra padding on both sides of y axis.
- (c) `enlarge x limits={upper=0.2}` adds 20% extra padding only on right sides of x axis. More options are `enlarge x limits={lower=<value>}`.



1.21 Coupled Harmonic Oscillators

From the knowledge of classical mechanics it is very well known the solution of positions of two particles :

$$x_1(t) = A_1 \cos(\omega_1 t + \phi_1) + A_2 \cos(\omega_2 t + \phi_2) \quad (1.19a)$$

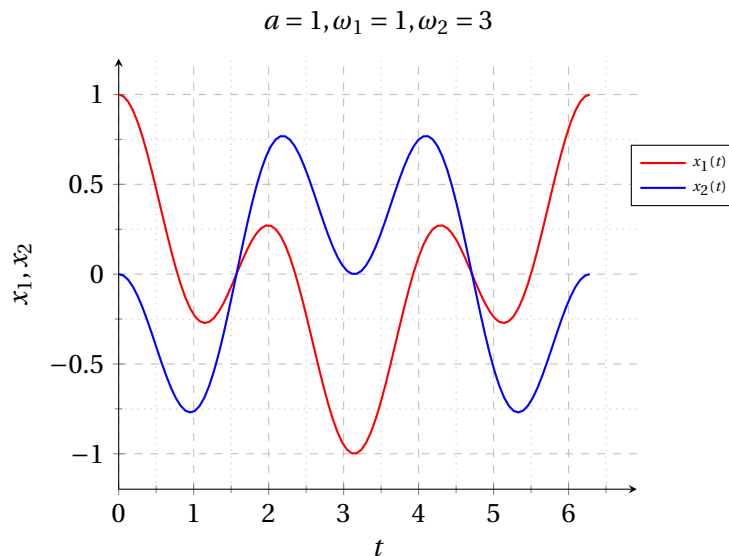
$$x_2(t) = A_1 \cos(\omega_1 t + \phi_1) - A_2 \cos(\omega_2 t + \phi_2) \quad (1.19b)$$

Where $\omega_1 = \sqrt{\frac{k}{m}}$ and $\omega_2 = \sqrt{\frac{k+2k'}{m}}$. Now suppose initially at $t = 0$ we have $x_1 = a$ and $x_2 = 0$ and both starts from rest, i.e. $\dot{x}_1(t=0) = \dot{x}_2(t=0) = 0$. Then we have the solution

$$x_1(t) = a \cos\left(\frac{\omega_1 + \omega_2}{2} t\right) \cos\left(\frac{\omega_1 - \omega_2}{2} t\right) \quad (1.20a)$$

$$x_2(t) = a \sin\left(\frac{\omega_1 + \omega_2}{2} t\right) \sin\left(\frac{\omega_1 - \omega_2}{2} t\right) \quad (1.20b)$$

Now the first oscillator stops and the second reaches maximum amplitude after when $\cos\left(\frac{\omega_1 - \omega_2}{2} t\right) = 0$. This produces periodic energy transfer (beats) between the oscillators as the energy of one oscillator is being transferred to another and vice-versa. We will show it very nicely by drawing the positions.



The following code was used to generate the above figure.

```

1 \begin{center}
2 \begin{tikzpicture}
3 % define amplitude
4 \pgfmathsetmacro{\a}{1} % $a$
5
6 % define frequencies
7 \pgfmathsetmacro{\omegaone}{1} % $\omega_1$
8 \pgfmathsetmacro{\omegatwo}{3} % $\omega_2$
9
10 \begin{axis}[
11 xlabel=$t$,ylabel={$x_1,x_2$},
12 grid=both,major grid style={dashed},
13 minor grid style={dotted},minor tick num=1,
14 axis lines=left,title={$a=\a,\omega_1=\omegaone,\omega_2=\omegatwo$},
15 legend style={at={(1.2,0.8)},font=\tiny},
16 enlarge x limits={upper=0.2},
17 enlarge y limits=0.1]
18
19 % define  $(\frac{\omega_1+\omega_2}{2})t$  and  $(\frac{\omega_1-\omega_2}{2})t$ 
20 \pgfmathdeclarefunction{w1pw2}{1}{%
21 \pgfmathparse{0.5*(\omegaone+\omegatwo)*#1}}
22
23 \pgfmathdeclarefunction{w1mw2}{1}{%
24 \pgfmathparse{0.5*(\omegaone-\omegatwo)*#1}}
25
26 % define  $x_1$ 
27 \pgfmathdeclarefunction{xone}{1}{%
28 \pgfmathparse{\a*cos(deg(w1pw2(#1)))*cos(deg(w1mw2(#1)))}}
29
30 % define  $x_2$ 
31 \pgfmathdeclarefunction{xtwo}{1}{%
32 \pgfmathparse{\a*sin(deg(w1pw2(#1)))*sin(deg(w1mw2(#1)))}}
33
34 % plot  $x_1$  and label it
35 \addplot[domain=0:2*pi,samples=100,red,thick]{xone(x)};
36 \addlegendentry{$x_1(t)$}
37
38 % plot  $x_2$  and label it
39 \addplot[domain=0:2*pi,samples=100,blue,thick]{xtwo(x)};
40 \addlegendentry{$x_2(t)$}
41 \end{axis}
42 \end{tikzpicture}
43 \end{center}

```

1.22 Superposition of Waves, Beats

The superposition of waves is the principle that when two or more waves overlap in space, the resulting displacement is the sum of the individual displacements at that point. This principle is fundamental in Wave Mechanics and Physics. Consider two waves $y_1(t)$ and $y_2(t)$ are superimposed on each other. The resultant would be $y(t) = y_1(t) + y_2(t)$. We take the following form :

$$y_1 = A \sin(\omega_1 t) \quad (1.21a)$$

$$y_2 = A \sin(\omega_2 t) \quad (1.21b)$$

The resultant wave is given by :

$$y = A \cos\left(\frac{\omega_1 - \omega_2}{2} t\right) \sin\left(\frac{\omega_1 + \omega_2}{2} t\right) \quad (1.22)$$

As we know $\sin\left(\frac{\omega_1 + \omega_2}{2} t\right)$ represents the rapid oscillation and $A \cos\left(\frac{\omega_1 - \omega_2}{2} t\right)$ represents the slowly varying amplitude. So the amplitude periodically increases and decreases. Lets plot this.

```

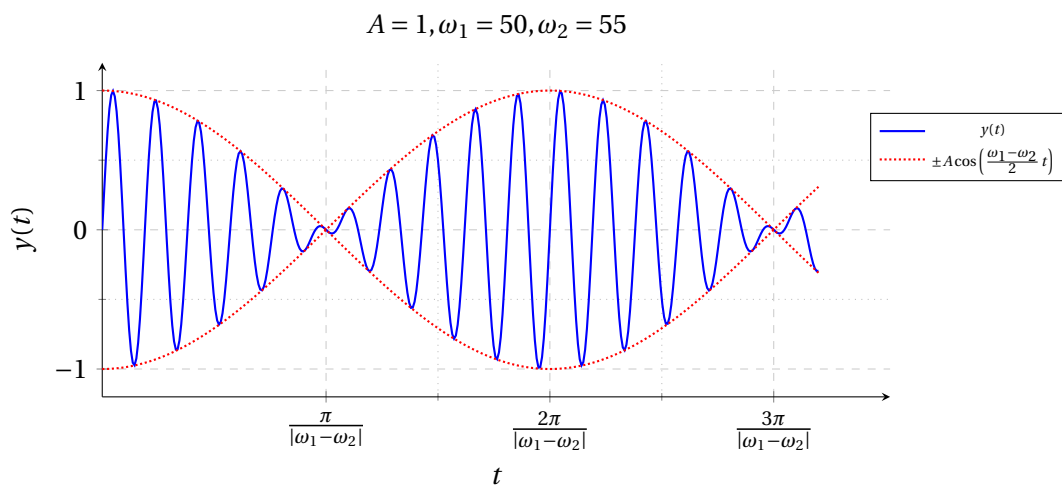
1 \begin{center}
2 \begin{tikzpicture}
3 % define amplitude
4 \pgfmathsetmacro{\A}{1} % $A$
5
6 % define frequencies
7 \pgfmathsetmacro{\omegaone}{50} % $\omega_1$
8 \pgfmathsetmacro{\omegatwo}{55} % $\omega_2$
9
10 % define the intervals
11 \pgfmathsetmacro{\ts}{pi/abs(\omegaone-\omegatwo)}
12 \begin{axis}[
13 xlabel=$t$,ylabel=$y(t)$,
14 grid=both,major grid style={dashed,gray!40},
15 minor grid style={dotted},minor tick num=1,
16 xtick={\ts,2*\ts,3*\ts},xticklabels={$\frac{\pi}{\abs{\omega_1-\omega_2}}$,
17 $\frac{2\pi}{\abs{\omega_1-\omega_2}}$, $\frac{3\pi}{\abs{\omega_1-\omega_2}}$},
18 xlabel style={yshift=0pt},
19 axis lines=left,domain=0:3.2*\ts,samples=400,width=12cm,height=6cm,
20 title={$A=\A$, $\omega_1=\omegaone$, $\omega_2=\omegatwo$},
21 legend style={at={(1.1,0.85)},font=\tiny,anchor=north},
22 enlarge x limits={upper=0.2},
23 enlarge y limits=0.1]
24
25 % define (\frac{\omega_1+\omega_2}{2} t and (\frac{\omega_1-\omega_2}{2} t
26 \pgfmathdeclarefunction{w1pw2}{1}{%
27 \pgfmathparse{0.5*(\omegaone+\omegatwo)*#1}}
28
29 \pgfmathdeclarefunction{w1mw2}{1}{%
30 \pgfmathparse{0.5*(\omegaone-\omegatwo)*#1}}
31
32 % define the net amplitude
33 \pgfmathdeclarefunction{Anet}{1}{%

```

```

34 \pgfmathparse{\A*cos(deg((w1mw2(#1))))}
35
36 % define y
37 \pgfmathdeclarefunction{ynet}{1}{%
38 \pgfmathparse{Anet(#1)*sin(deg((w1pw2(#1))))}}
39
40 % plot y and label it
41 \addplot[blue,thick]{ynet(x)};
42 \addlegendentry{\$y(t)\$}
43
44 % plot envelopes
45 \addplot[red,densely dotted,thick]{Anet(x)};
46 \addplot[red,densely dotted,thick]{-Anet(x)};
47 \addlegendentry{\$ \pm A \cos \qty(\frac{\omega_1 - \omega_2}{2} t) \$}
48
49 \end{axis}
50 \end{tikzpicture}
51 \end{center}

```



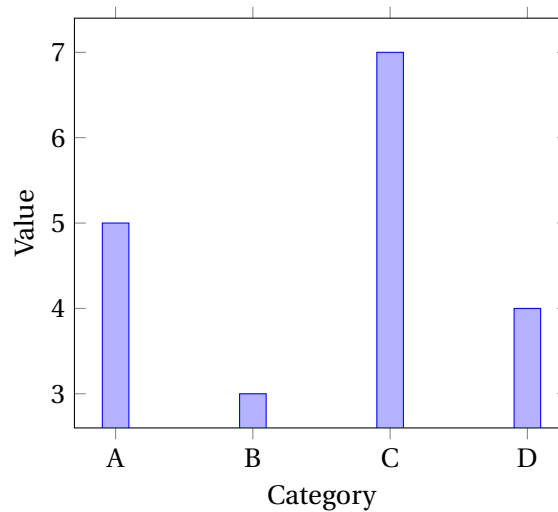
1.23 Bar plots

Bar plots are useful for representing discrete data such as categories, counts, or comparisons. PGFPlots provides built-in styles like `ybar` and `xbar` to create vertical and horizontal bar charts. Here is a minimal example.

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[ybar,xlabel={Category},ylabel={Value},
4 symbolic x coords={A,B,C,D},xtick=data,height=7cm]
5
6 \addplot coordinates {(A,5) (B,3) (C,7) (D,4)};
7 \end{axis}
8 \end{tikzpicture}
9 \end{center}

```



Let's explain :

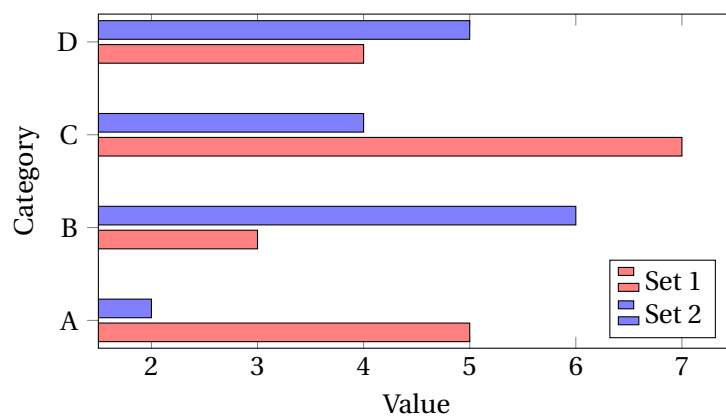
- (a) `ybar` draws vertical bars.
- (b) `symbolic x coords` is used to label the category.
- (c) `xtick=data` places ticks at each category.
- (d) `(A, 5)` means category A has value 5.
- (e) If you want to draw horizontal bars use `ybar`, `symbolic x coords={...}`, `ytick`.

Here is an example of multiple bar plots in the same figure.

```

1  \begin{center}
2  \begin{tikzpicture}
3  \begin{axis}[xbar,bar width=7pt,ylabel={Category},
4  xlabel={Value},symbolic y coords={A,B,C,D},
5  ytick=data,legend pos=south east,height=6cm,width=10cm]
6
7  \addplot[fill=red!50] coordinates {(5,A) (3,B) (7,C) (4,D)};
8  \addplot[fill=blue!50] coordinates {(2,A) (6,B) (4,C) (5,D)};
9  \legend{Set 1, Set 2}
10
11 \end{axis}
12 \end{tikzpicture}
13 \end{center}

```



- (a) The command `bar width=10pt` sets the bar width to 10pt.
- (b) `\addplot[fill=red!50]` fills the bar with 50% red colour.

1.24 Drawing Wavefunctions

Consider a quantum mechanical particle with mass m is confined in a “Infinite Potential Well” of length L . We know the energy states are discrete. They are given by :

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad (1.23)$$

And the wave functions are given by :

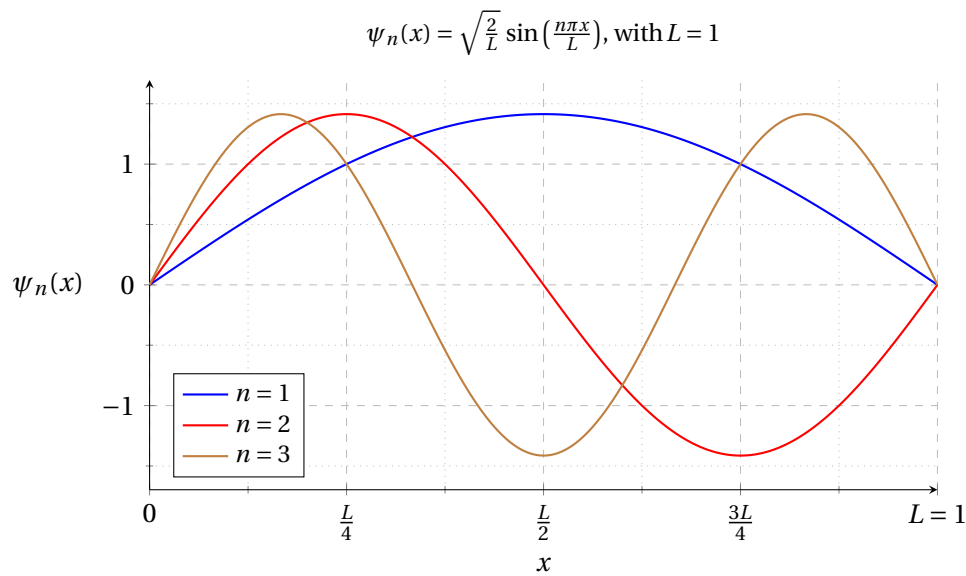
$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad (1.24)$$

Where $n = 1, 2, \dots$. We now focus our attention to plot the wavefunctions, only few of them.

```

1  \begin{center}
2  \begin{tikzpicture}
3
4  % define the Length of the box
5  \pgfmathsetmacro{\L}{1} % length of the well
6
7  % define normalization constant
8  \pgfmathsetmacro{\A}{sqrt(2/\L)}
9
10 % define wavefunction
11 \pgfmathdeclarefunction{psi}{2}{%
12 \pgfmathparse{\A*sin(deg(#1*pi*#2/\L))}}
13
14 \begin{axis}[xlabel=$x$,ylabel=$\psi_n(x)$, ylabel style={rotate=270},
15 title={$\psi_n(x)=\sqrt{\frac{2}{L}}\sin(\frac{n\pi x}{L})$, \text{with} L=\L$},
16 title style={font=\small}, domain=0:\L, samples=200, grid=both, legend pos = south west,
17 major grid style={dashed}, minor grid style={dotted}, minor tick num =1,
18 legend style={font=\small}, axis lines=left, xtick={0,0.25*\L,0.5*\L,0.75*\L,\L},
19 xticklabels={$0$, $\frac{L}{4}$, $\frac{L}{2}$, $\frac{3L}{4}$, $\L=\L$},
20 enlarge y limits = 0.1,height=7cm,width=12cm]
21
22 \addplot[blue,thick]{psi(1,x)};
23 \addlegendentry{$n=1$}
24
25 \addplot[red,thick]{psi(2,x)};
26 \addlegendentry{$n=2$}
27
28 \addplot[brown,thick]{psi(3,x)};
29 \addlegendentry{$n=3$}
30 \end{axis}
31 \end{tikzpicture}
32 \end{center}

```



1.25 Beyond 2D, Plotting 3D surfaces

PGFPlots can render smooth 3D surfaces, meshes, contour maps, and parametric surfaces directly in LaTeX. The basic environment remains the same:

```

1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[options]
4 \addplot3[options] {function};
5 \end{axis}
6 \end{tikzpicture}
7 \end{center}

```

`\addplot3` is the command that takes two arguments (x, y) and plots the surface. Let's plot a function

$$z = \sin x \cos y \quad (1.25)$$

```

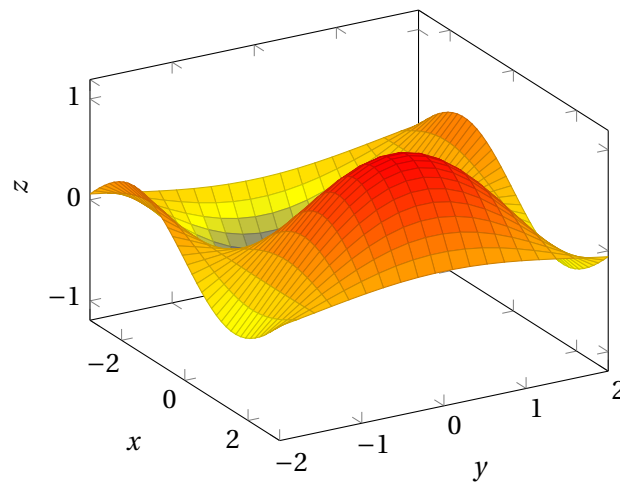
1 \begin{center}
2 \begin{tikzpicture}
3 \begin{axis}[view={60}{30}, xlabel={\$x\$}, ylabel={\$y\$},
4 zlabel={\$z\$}, samples=40, samples y=20]
5
6 \addplot3[surf, domain=-3:3, domain y=-2:2] {\sin(deg(x))*\cos(deg(y))};
7 \end{axis}
8 \end{tikzpicture}
9 \end{center}

```

(a) `view={60}{30}` sets the viewing angle. The basic syntax is `view={azimuth}{elevation}`.

(b) `samples=40, samples y=20` sets the no of samples (40,20) along x and y respectively.

(c) `\addplot3[surf, domain=-3:3, domain y=-2:2]` here makes a surface, where $-3 \leq x \leq 3$ and $-2 \leq y \leq 2$.



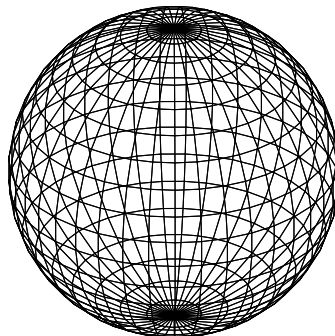
We can also define surfaces parametrically. For example plotting a sphere we take the following form :

$$x = \sin u \cos v$$

$$y = \sin u \sin v$$

$$z = \cos u$$

Where $0 \leq u \leq \pi$ and $0 \leq v \leq 2\pi$.



```

1  \begin{center}
2  \begin{tikzpicture}[scale=2]
3  \begin{axis}[samples=40,samples y=20,
4  domain=0:180,domain y=0:360,
5  axis lines=none,axis equal]
6
7  \addplot3[mesh,black,very thin]({sin(x)*cos(y)},{sin(x)*sin(y)},{cos(x)});
8  \end{axis}
9  \end{tikzpicture}
10 \end{center}

```

1. `\begin{tikzpicture}[scale=2]` scales the whole figure by a factor of 2.
2. `axis lines=none` removes all the axes and `axis equal` ensures that all the axes have same scaling.
3. `\addplot3[mesh,black,very thin]`, mesh command here produces a wire frame grid that is black coloured with very thin line-style.

1.26 2D Infinite Potential Well

In this case wave functions are given by :

$$\psi_{n_1, n_2}(x, y) = \sqrt{\frac{4}{L_x L_y}} \sin\left(\frac{n_1 \pi x}{L_x}\right) \sin\left(\frac{n_2 \pi y}{L_y}\right) \quad (1.26)$$

And the probability density is given by :

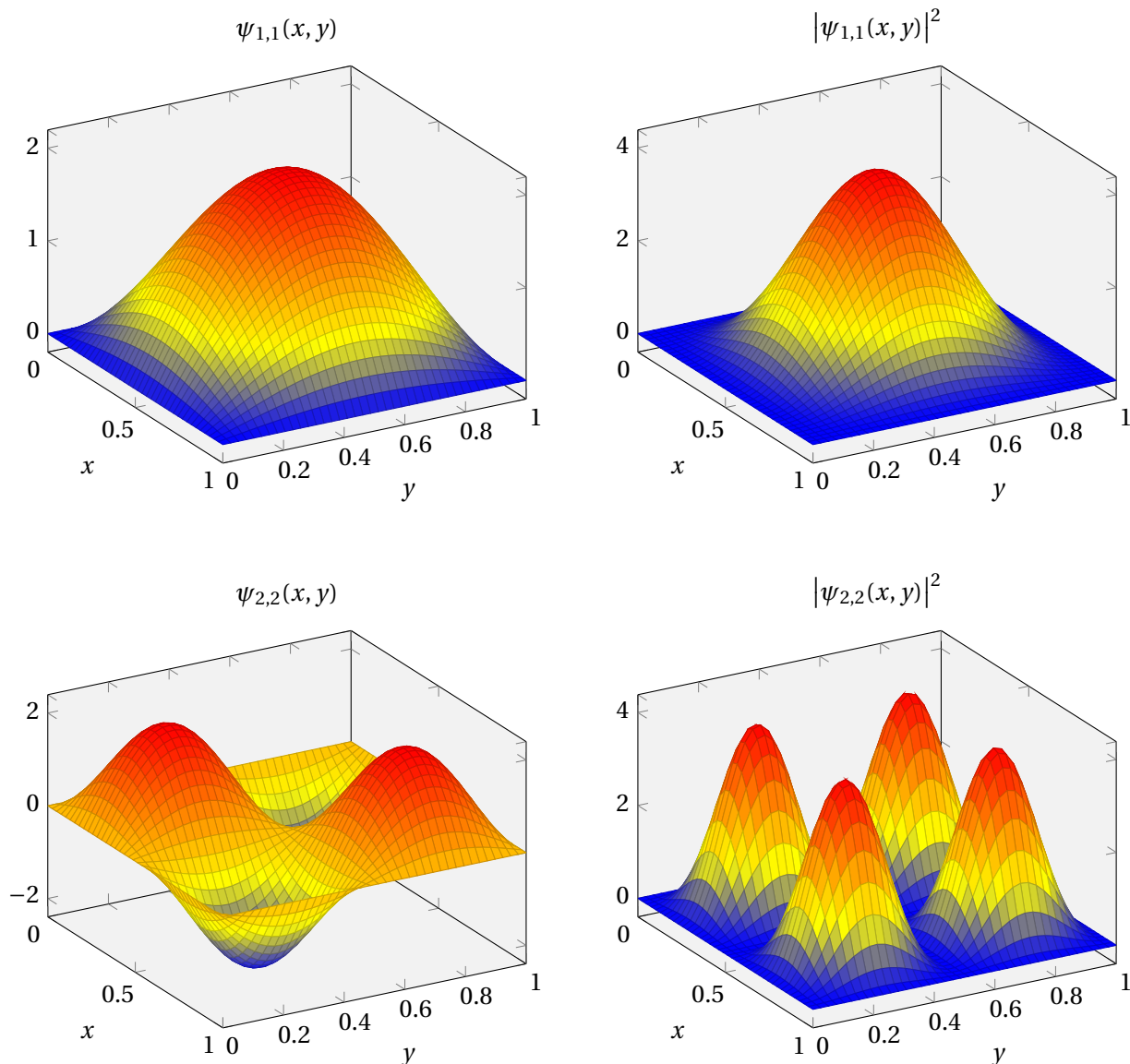
$$P(x, y) = |\psi_{n_1, n_2}(x, y)|^2 \quad (1.27)$$

Where $n_1, n_2 = 1, 2, \dots$. We now focus our attention to plot the wavefunctions and the corresponding probability density.

```

1  \begin{center}
2  \begin{tikzpicture}
3  % define the Length of the box
4  \pgfmathsetmacro{\Lx}{1} % length of the well along x
5  \pgfmathsetmacro{\Ly}{1} % length of the well along y
6
7  % define wavefunction along 1D
8  \pgfmathdeclarefunction{psix}{3}{%
9  \pgfmathparse{sqrt(2/#1)*sin(deg(#2*pi*#3/#1))}}
10
11 % define wavefunction
12 \pgfmathdeclarefunction{psi}{6}{%
13 \pgfmathparse{psix(#1,#2,#3)*psix(#4,#5,#6)}}
14
15 \begin{groupplot}[group style={group size=2 by 2, horizontal sep=16mm, vertical sep=24mm},
16 view={60}{30}, xlabel={\$x\$}, ylabel={\$y\$}, zlabel={\$z\$}, domain=0:\Lx, domain y=0:\Ly,
17 samples=40, samples y=40, axis background/.style={fill=gray!10}]
18
19 \nextgroupplot[title={\$ \psi_{1,1}(x,y) \$}] % plot n1=n2=1 wavefunction
20 \addplot3[surf]{psi(\Lx,1,x,\Ly,1,y)};
21
22 \nextgroupplot[title={\$ \abs{\psi_{1,1}(x,y)}^2 \$}] % plot n1=n2=1 probability
23 \addplot3[surf]{psi(\Lx,1,x,\Ly,1,y)^2};
24
25 \nextgroupplot[title={\$ \psi_{2,2}(x,y) \$}] % plot n1=n2=2 wavefunction
26 \addplot3[surf]{psi(\Lx,2,x,\Ly,2,y)};
27
28 \nextgroupplot[title={\$ \abs{\psi_{2,2}(x,y)}^2 \$}] % plot n1=n2=2 probability
29 \addplot3[surf]{psi(\Lx,2,x,\Ly,2,y)^2};
30
31 \end{groupplot}
32 \end{tikzpicture}
33 \end{center}

```



1.27 Roses and Roses

In Sec. 1.18 we have discussed about the plots if the k is integer. Now we will allow them to be fractional also. Here is a generic code which does this.

```

1  %-----
2  % Generic rose macro
3  % Usage: \drawrose{a}{p}{q}{sin/cos} where k=p/q, p,q integers
4  %-----
5  \newcommand{\drawrose}[4]{%
6  % Convert k = p/q
7  \pgfmathsetmacro{\k}{#2/#3}%
8  % Compute full revolutions depending on numerator parity
9  \pgfmathtruncatemacro{\revs}{ifthenelse(mod(#2,2)==1,#3,2*#3)}%
10 % Draw the rose
11 \addplot[domain=0:\revs*360,samples=200,thin,blue]
12 ({#1*#4*(\k*x)*cos(x)}, {#1*#4*(\k*x)*sin(x)});
13

```

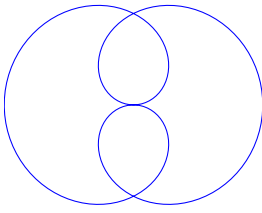
```

14 \begin{center}
15 \begin{tikzpicture}
16 \begin{groupplot}[group style={group size=5 by 5,horizontal sep=3mm,vertical sep=7mm},
17 axis lines=none,axis equal,width=5cm,height=6cm,xtick=\empty,ytick=\empty,
18 enlargelimits=false,title style={at={(0.5,0.9)}}]
19 % Row 1
20 \nextgroupplot[title={$k=\frac{1}{2}$}]\drawrose{1}{1}{2}{sin}
21 \nextgroupplot[title={$k=\frac{2}{3}$}]\drawrose{1}{2}{3}{cos}
22 \nextgroupplot[title={$k=\frac{3}{4}$}]\drawrose{1}{3}{4}{sin}
23 \nextgroupplot[title={$k=\frac{4}{5}$}]\drawrose{1}{4}{5}{cos}
24 \nextgroupplot[title={$k=\frac{5}{6}$}]\drawrose{1}{5}{6}{sin}
25 % Row 2
26 \nextgroupplot[title={$k=\frac{6}{7}$}]\drawrose{1}{6}{7}{cos}
27 \nextgroupplot[title={$k=\frac{7}{8}$}]\drawrose{1}{7}{8}{sin}
28 \nextgroupplot[title={$k=\frac{8}{9}$}]\drawrose{1}{8}{9}{cos}
29 \nextgroupplot[title={$k=\frac{9}{10}$}]\drawrose{1}{9}{10}{sin}
30 \nextgroupplot[title={$k=\frac{10}{11}$}]\drawrose{1}{10}{11}{cos}
31 % Row 3
32 \nextgroupplot[title={$k=\frac{11}{12}$}]\drawrose{1}{11}{12}{sin}
33 \nextgroupplot[title={$k=\frac{12}{13}$}]\drawrose{1}{12}{13}{cos}
34 \nextgroupplot[title={$k=\frac{13}{14}$}]\drawrose{1}{13}{14}{sin}
35 \nextgroupplot[title={$k=\frac{14}{15}$}]\drawrose{1}{14}{15}{cos}
36 \nextgroupplot[title={$k=\frac{15}{16}$}]\drawrose{1}{15}{16}{sin}
37 % Row 4
38 \nextgroupplot[title={$k=\frac{16}{17}$}]\drawrose{1}{16}{17}{cos}
39 \nextgroupplot[title={$k=\frac{17}{18}$}]\drawrose{1}{17}{18}{sin}
40 \nextgroupplot[title={$k=\frac{18}{19}$}]\drawrose{1}{18}{19}{cos}
41 \nextgroupplot[title={$k=\frac{19}{20}$}]\drawrose{1}{19}{20}{sin}
42 \nextgroupplot[title={$k=\frac{20}{21}$}]\drawrose{1}{20}{21}{cos}
43 % Row 5
44 \nextgroupplot[title={$k=\frac{21}{22}$}]\drawrose{1}{21}{22}{sin}
45 \nextgroupplot[title={$k=\frac{22}{23}$}]\drawrose{1}{22}{23}{cos}
46 \nextgroupplot[title={$k=\frac{23}{24}$}]\drawrose{1}{23}{24}{sin}
47 \nextgroupplot[title={$k=\frac{24}{25}$}]\drawrose{1}{24}{25}{cos}
48 \nextgroupplot[title={$k=\frac{25}{26}$}]\drawrose{1}{25}{26}{sin}
49 \end{groupplot}
50 \end{tikzpicture}
51 \end{center}

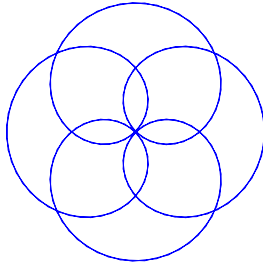
```

1. `\pgfmathttruncatemacro` truncates a number to its nearest integer.
2. `mod(#2,2)==1` computes the remainder of #2 divided by 2 and checks if it is 1 (i.e., if the number is odd).
3. `\ifthenelse(<condition>, <true case>, <false case>)` is the syntax for conditional expressions. So `\ifthenelse(mod(#2,2)==1,#3,2*#3)` returns #3 if #2 is odd, and $2 \cdot \#3$ if #2 is even.
4. `enlargelimits=false` removes any extra space around axes. We plotted sin and cosine alternatively.

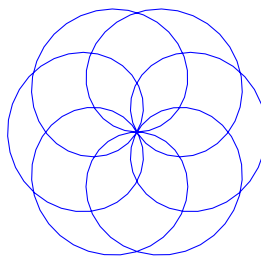
$$k = \frac{1}{2}$$



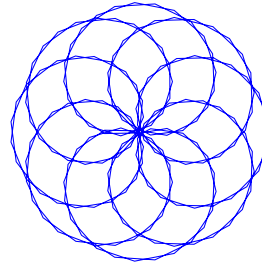
$$k = \frac{2}{3}$$



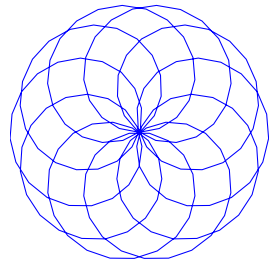
$$k = \frac{3}{4}$$



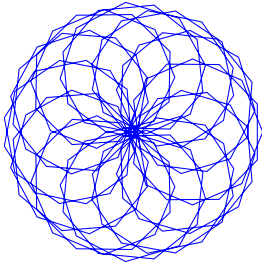
$$k = \frac{4}{5}$$



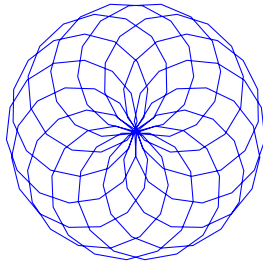
$$k = \frac{5}{6}$$



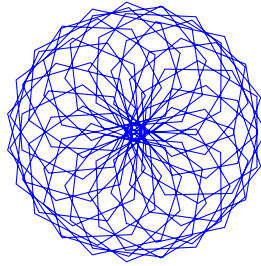
$$k = \frac{6}{7}$$



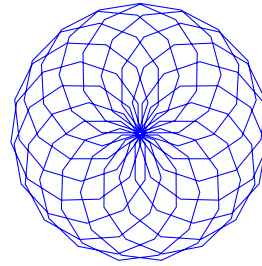
$$k = \frac{7}{8}$$



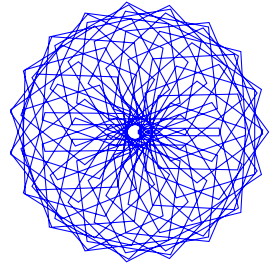
$$k = \frac{8}{9}$$



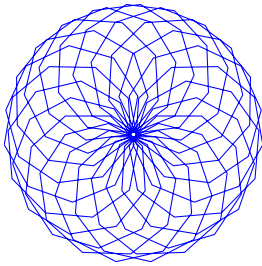
$$k = \frac{9}{10}$$



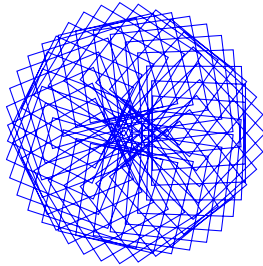
$$k = \frac{10}{11}$$



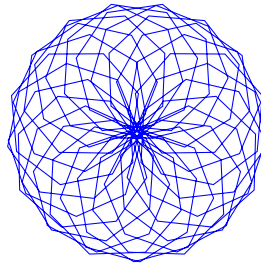
$$k = \frac{11}{12}$$



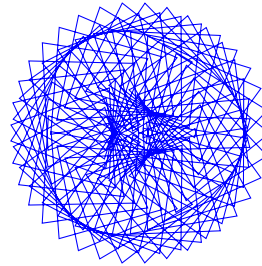
$$k = \frac{12}{13}$$



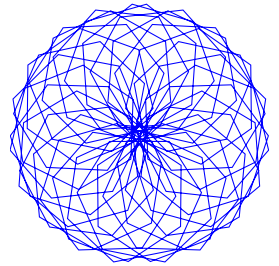
$$k = \frac{13}{14}$$



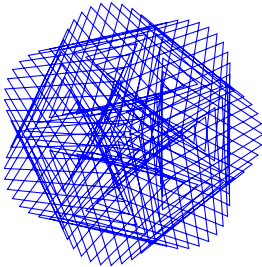
$$k = \frac{14}{15}$$



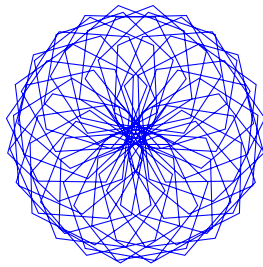
$$k = \frac{15}{16}$$



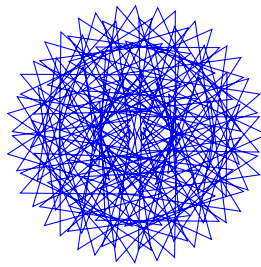
$$k = \frac{16}{17}$$



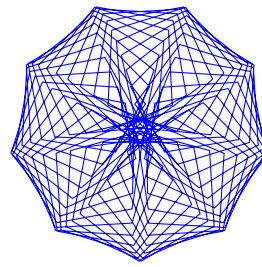
$$k = \frac{17}{18}$$



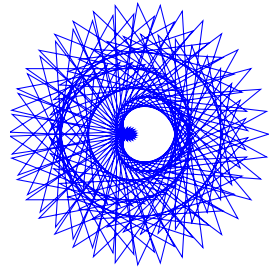
$$k = \frac{18}{19}$$



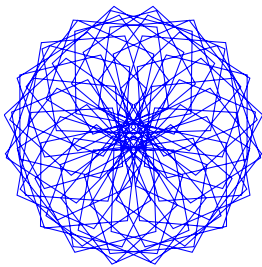
$$k = \frac{19}{20}$$



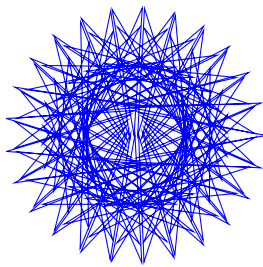
$$k = \frac{20}{21}$$



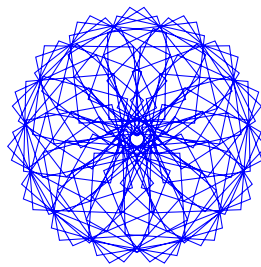
$$k = \frac{21}{22}$$



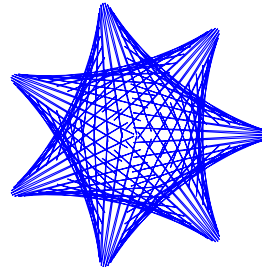
$$k = \frac{22}{23}$$



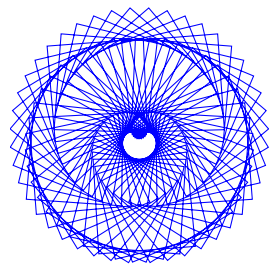
$$k = \frac{23}{24}$$



$$k = \frac{24}{25}$$



$$k = \frac{25}{26}$$



1.28 Single Slit Diffraction

Consider a slit of width a illuminated by monochromatic light of wavelength λ . The slit is perpendicular to the incident plane wave, and we observe the diffraction pattern on a screen far away (Fraunhofer diffraction). From our knowledge of optics we know, the intensity on the screen is proportional to the square of the amplitude which is given by :

$$I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \quad (1.28)$$

Where $\beta = \frac{\pi a \sin \theta}{\lambda}$. I_0 is the intensity at $\theta = 0$, the max intensity also known as central maximum. The dark fringes occur when the numerator vanishes:

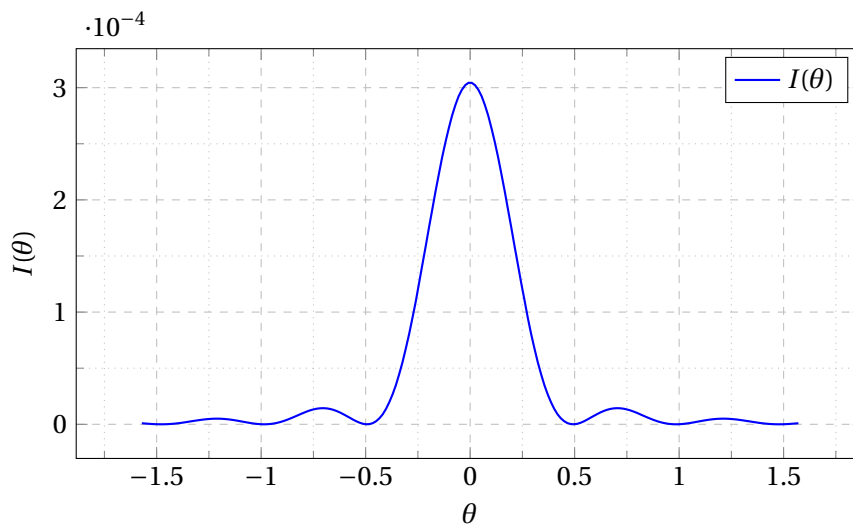
$$\beta = m\pi \Rightarrow a \sin \theta = m\lambda \quad m = \pm 1, \pm 2, \dots \quad (1.29)$$

These are the angles for the minima. Let us plot the Intensity pattern to have a better visualization.

```

1  \begin{center}
2  \begin{tikzpicture}
3  \begin{axis}[xlabel=$\theta$,ylabel=$I(\theta)$,height=7cm,width=12cm,
4  grid=both,major grid style={dashed},minor grid style={dotted},minor tick num=1]
5
6  % slit parameters, wavelength and max intensity
7  \pgfmathsetmacro{\a}{0.004}           % slit width in meters
8  \pgfmathsetmacro{\wl}{0.0000006}     % wavelength in meters
9  \pgfmathsetmacro{\Inot}{1}           % max intensity
10
11 % define beta
12 \pgfmathdeclarefunction{bta}{1}{%
13 \pgfmathparse{pi*\a*sin(#1)/\wl}}
14
15 % Define I(theta)
16 \pgfmathdeclarefunction{It}{1}{%
17 \pgfmathparse{\Inot*((sin(bta(#1))/bta(#1))^2)}}
18
19 \addplot[domain=-pi/2:pi/2,samples=200,blue,thick]{It(x)};
20 \addlegendentry{$I(\theta)$}
21 \end{axis}
22 \end{tikzpicture}
23 \end{center}

```



Here we have taken the slit width $a = 4\text{mm}$, $\lambda = 600\text{nm}$ and $I_0 = 1$.

1.29 Type Setting Notes

- (a) This document has been produced using $\text{\LaTeX}$ typesetting.
- (b) The font used is *fourier*. To use this font, include the line `\usepackage{fourier}` in the preamble.
- (c) All the figures have been generated using the `TikZ` and `PGFPlots` package.
- (d) The $\text{\LaTeX}$ source for this document can be provided by the corresponding author upon reasonable request.